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REVIEW

Microscopic mechanisms of oil-water-rock interactions during CO_2 enhanced oil recovery and storage: a comprehensive review

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Abstract

With the large-scale deployment of carbon capture, utilization, and storage (CCUS) technologies, CO_2-enhanced oil recovery and storage (CO_2-EOR & Storage) has emerged as a core technical approach that improves crude oil recovery efficiency and enables permanent carbon sequestration. The operational efficiency and long-term stability of this technology rely heavily on microscopic interactions occurring within reservoir oil-water-rock multiphase systems. This paper systematically investigates the physical interactions and chemical reaction mechanisms between injected CO_2 and reservoir crude oil, formation water, and rock media. After reservoir injection, CO_2 dissolves into crude oil and triggers oil swelling, viscosity decline, light component extraction, and miscible phase formation, which modifies crude oil fluidity at the molecular scale. CO_2 also dissolves in formation water to generate weak carbonic acid. The resultant acid reacts with rock minerals through dissolution, precipitation, and ion exchange and alters reservoir wettability, which dynamically reorganizes pore-throat geometries and subsurface flow environments. Competitive adsorption at oil–water–rock triple-phase interfaces, combined with the balance between capillary and viscous forces as well as multicomponent mass transfer and diffusion, governs the entire spectrum of microscopic interactions within the studied system. Existing research has transitioned from macroscopic characterization to nanoscale mechanistic investigations. Unresolved issues remain regarding reaction pathways under multiphase and multi-field coupling conditions, dynamic interfacial behaviors, and quantitative relationships between microscopic mechanisms and macroscopic engineering performances. This review elaborates the microscopic interaction mechanisms of CO_2 with oil-water-rock systems, delivering universal theoretical implications for reservoir research worldwide. It provides fundamental theoretical support for the optimization of global CO_2-EOR and carbon sequestration projects and offers generalized theoretical guidance for international CCUS deployment.

Keywords CO_2 flooding · Carbon storage · Oil–water–rock · Microscopic reaction · Chemical reaction · CCUS · Enhanced oil recovery

1 Introduction

Driven by global carbon neutrality initiatives and widespread carbon capture, utilization, and storage (CCUS) deployment, CO_2-enhanced oil recovery and storage (CO_2-EOR & Storage) has become a viable technical solution that yields two key outcomes: improved crude oil recovery and permanent geological carbon sequestration. This technique bridges national energy security demands and deep decarbonization targets through practical engineering applications. The performance and operational stability of field-scale CO_2-EOR and storage operations are governed by microscale interactions within reservoir oil–water–rock multiphase systems. These intricate microscopic reactions control CO_2 dissolution, diffusion, and migration behaviors, while dictating crude oil flow characteristics, rock property evolution, and long-term carbon sequestration security. Microscopic reaction mechanisms remain a key scientific constraint limiting the large-scale industrial adoption of this technology [1]. Clarification of the inherent principles and dynamic evolution characteristics of in-reservoir CO_2 microscopic reaction mechanisms underpins the scale-up of

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CO_2 flooding and sequestration technologies from laboratory investigations to field-scale oilfield implementation. Microscopic CO_2–crude oil interactions are dominated by dissolution, oil swelling, viscosity reduction, light component extraction, and miscible phase formation. These physical behaviors lower interfacial tension and enhance fluid mobility to achieve efficient residual oil displacement [2, 3]. CO_2 dissolution in formation water generates weak carbonic acid, which reacts with rock minerals via dissolution, precipitation, ion exchange, and wettability alteration. Such reactions restructure pore-throat geometries and modify subsurface flow conditions [4, 5]. Competitive adsorption at oil–water–rock triple-phase interfaces, dynamic equilibrium between capillary and viscous forces, and multicomponent interphase mass transfer and diffusion jointly shape the full evolutionary framework of in-reservoir CO_2 multiphase microscopic interactions [6, 7]. This framework can be categorized into three research dimensions. First, for oil phase systems, CO_2 dissolution restructures crude oil molecular configurations at the molecular scale. Variations in crude oil density, viscosity, and diffusion coefficient initiate convective mixing and density fingering. The microscopic kinetic features of these processes dominate oil displacement performance [2]. Shifts in CO_2–crude oil miscibility alter oil phase viscosity and interfacial tension, which modulates fluid flow resistance within pore throats [8]. Second, for water phase systems, CO_2–formation water interactions induce aqueous acidification and ionic rearrangement, which modulates rock surface wettability and pore architecture [9]. Carbonic acid generated from CO_2–water reactions drives water–rock interfacial dissolution and precipitation. The resultant changes in rock pore structure and mineral composition can optimize flow conduits or trigger scale-induced reservoir damage [10]. Third, for rock phase systems, carbonic acid derived from dissolved CO_2 triggers mineral dissolution, precipitation, and ion exchange, reshaping the geometry and connectivity of reservoir pore throats [11]. Mineral chemical reactions, clay swelling and migration, and dynamic pore structure evolution affect long-term reservoir stability and determine the availability of flow pathways and carbon sequestration voids [12]. Research focus has transitioned from macroscopic displacement rules to nanoscale mechanistic exploration. Systematic collation and integration are still required to clarify multiphase and multi-field coupled reaction pathways, dynamic interfacial evolution, and quantitative relationships between microscopic mechanisms and macroscopic engineering parameters. Microscopic interactions between CO_2 and reservoir media persist throughout the full displacement and carbon sequestration lifecycle [2]. Reservoirs represent complex multiphase and multicomponent geological systems. CO_2 injection disturbs the inherent thermodynamic equilibrium of oil–water–rock systems and induces a series of microscale physicochemical variations in subsurface formations [13]. This paper reviews physicochemical coupling mechanisms in the CO_2–oil–water–rock multiphase multiphysics system. It elucidates the integrated microscale reaction framework encompassing oil-phase fluid modification, water-phase acidification, and dynamic evolution of rock pore throats, and examines how multiscale microscopic behavior controls oil displacement efficiency, reservoir properties, and geological sequestration safety. The work refines the fundamental theoretical framework for CO_2-enhanced oil recovery (CO_2-EOR) and geological sequestration. It supports laboratory investigations including molecular simulations and core-based physical modeling, and links microscopic mechanisms to field development practices. This linkage provides a theoretical basis for optimizing injection strategies, mitigating reservoir damage, and designing long-term carbon sequestration schemes.

Existing research on these microscale mechanisms remains fragmented, with no unified theoretical framework or integrated schematic that intuitively captures multiphase and multi-field coupling relationships. This paper presents a conceptual diagram of microscale coupling mechanisms for CO_2 flooding and sequestration (Fig. 1). The diagram categorizes and integrates core reactions across three subsystems: CO_2–crude oil, CO_2–formation water, and CO_2–rock. It characterizes spatiotemporal evolution patterns of microscale interactions during CO_2 front migration and their cross-subsystem coupling relationships, forming a visualized, integrated framework for theoretical analysis.

2 Current research status on the microkinetic reaction mechanism of CO_2–oil systems

CO_2 flooding is a key enhanced oil recovery technique, the degree to which its microscopic mechanisms are understood determines the scientific validity of engineering design and the effectiveness of field implementation. When CO_2 dissolves in crude oil, it weakens intermolecular forces, reduces viscosity, and promotes volume expansion, under supercritical conditions, its preferential extraction of light components further improves fluid mobility. In shale oil development, CO_2 exploits its greater adsorption affinity to displace adsorbed hydrocarbons, achieving synergistic oil displacement and carbon sequestration. The nanoregional effect influences the microscopic sweep efficiency of CO_2 flooding by enhancing or suppressing diffusive mass transfer. Researchers worldwide have examined four principal mechanisms—dissolution-induced swelling and viscosity reduction, phase extraction, competitive adsorption, and nanopore mass transfer—studies that underpin

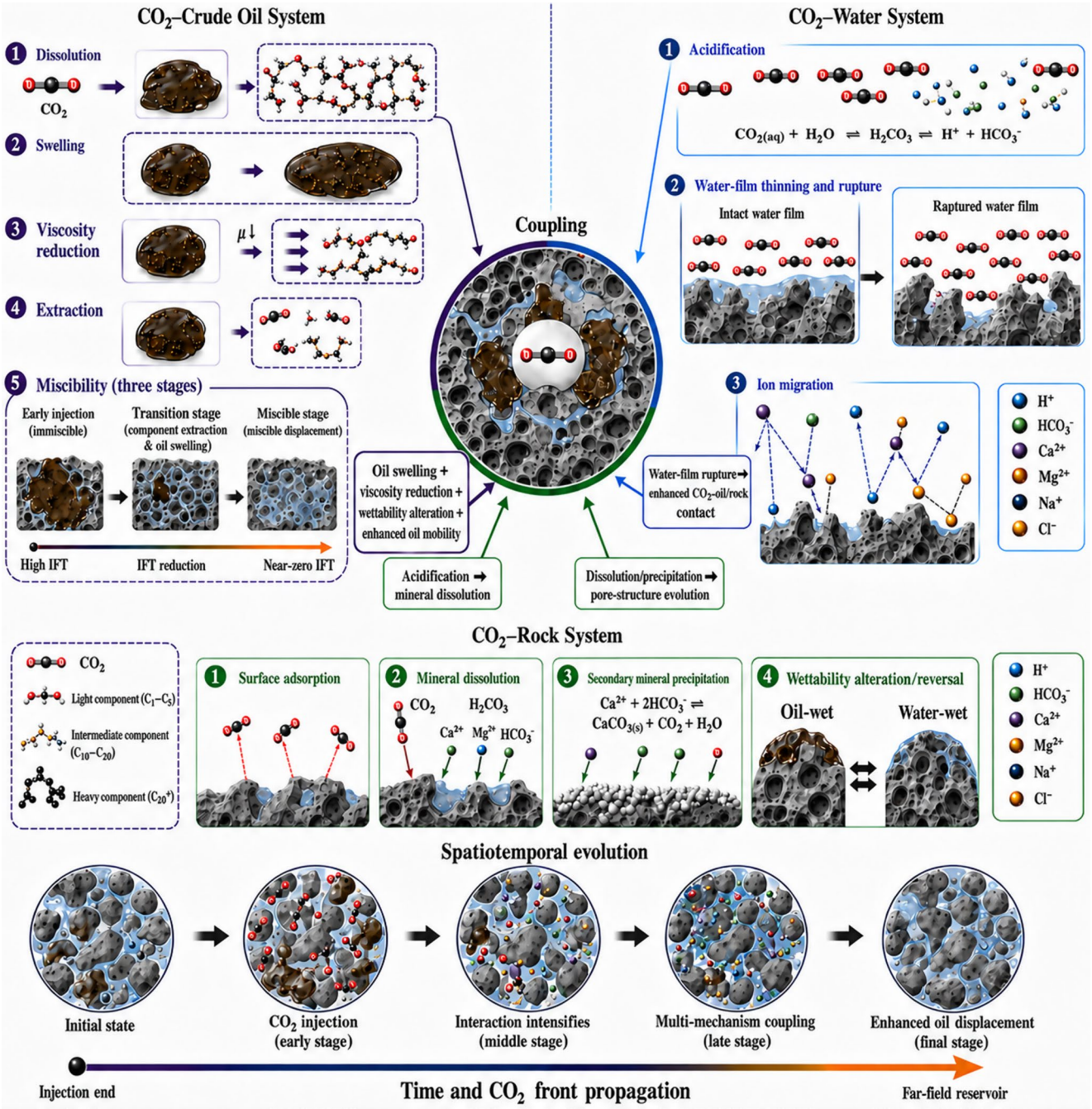


Fig. 1 Schematic diagram of multiphase coupling microscopic interactions and spatiotemporal evolution of CO_2-oil-water-rock system

understanding of the microscopic reactions in CO_2 flooding and inform efforts to improve oil displacement efficiency.

2.1 Dissolution–swelling–viscosity reduction mechanism

2.1.1 Solubilization and viscosity reduction mechanism

During CO_2-enhanced oil recovery (CO_2-EOR), the dissolution of CO_2 in crude oil is the primary physicochemical mechanism that triggers favorable oil displacement effects. As an active fluid, CO_2 dissolves extensively in the

hydrocarbon system upon contact with crude oil, inducing changes in physical properties—including reduced oil viscosity, volume expansion, density adjustment, and decreased oil–gas interfacial tension—whose combined effect directly improves oil displacement efficiency [14, 15]. Many studies have examined the mechanisms and key controlling factors governing CO_2 dissolution behavior from various perspectives.

Guo et al. (2018) performed PVT and slim-tube displacement experiments to examine CO_2 dissolution behavior in tight oil reservoirs of the Changteng Oilfield. CO_2 solubility rises with pressure and concentration, reflected in a higher bubble-point pressure. A higher bubble-point pressure signals greater CO_2 dissolution, which lowers crude-oil density and narrows the density contrast between the oil and gas phases. By the Laplace equation, this reduced density contrast decreases interfacial tension and thus the minimum miscibility pressure; a higher bubble-point pressure therefore enables miscibility at lower pressures, reducing residual oil saturation and improving oil displacement efficiency [16]. Gandomkar et al. (2024) showed experimentally that accurate determination of interfacial tension (IFT) relies on precise calibration of the oil–gas density difference ($\Delta\rho$). $\Delta\rho$ is measured by the mass–volume method in a high-temperature, high-pressure reactor, whereas gas-phase density can be predicted with the Peng–Robinson (PR) equation of state or obtained from property databases. When the gas phase contains polymer thickeners such as PFA, however, the mixture becomes complex, binary interaction coefficients are usually unavailable, and equation-of-state (EOS) deviations grow; in such cases, regression against experimentally measured data and rigorous correction for instrument volume expansion at high pressure are required to ensure reliable IFT fitting [17, 18]. Seyyedi (2018) compared CO_2 flooding with carbonated water flooding experimentally and analyzed differences in their microscale interactions with residual oil by tracking changes in its composition and viscosity. Pure CO_2 flooding raised residual oil viscosity from 82 cp. to 130 cp. (a 58% increase) and increased density to 0.932 g/cm³, shifting the composition sharply toward heavier fractions; carbonated water injection, by contrast, kept residual oil viscosity within 82–95 cp. and density stable at 0.923–0.928 g/cm³, markedly reducing residual oil viscosity [19]. Babaei et al. (2018) used molecular dynamics simulations to examine how molecular diffusion rates affect CO_2 flooding efficiency and carbon sequestration. Molecular diffusion can increase CO_2 solubility in crude oil and improve oil recovery; with diffusion enabled, CO_2 solubility in the oil phase rises sharply, and solubility differences across diffusion coefficient configurations can reach approximately 13%. Higher diffusion coefficients are not always preferable; when the CO_2 diffusion coefficient is excessively high—causing a corresponding drop in the diffusion coefficients of heavy components such as $C_{10}H_{22}$—reduced heavy-hydrocarbon mobility traps the CO_2-bearing oil phase in the pores, lowering oil recovery from approximately 94% to approximately 90%, a decline of about 4% points [20]. Zhang et al. (2023) performed high-pressure phase-behavior and asphaltene experiments on crude oil from well S13-95 in the North China Oilfield. Various concentrations of CO_2 were injected at 108 °C, followed by a saturation–depletion cycle, and the interaction between CO_2 and the crude oil was examined at the molecular level. CO_2 dissolution weakens the intermolecular (van der Waals) forces in the crude oil, causing its volume to expand by 45% and its viscosity and density to decrease by 64.2% and 16.4%, respectively, which improves fluid mobility [21]. Li et al. (2023) combined microscale experiments with numerical simulations to reveal the synergistic effect of advection—a macroscopic flow process in which the fluid moves under an external driving force and carries dissolved or mixed components with it—and diffusion on the recovery factor during CO_2 flooding. In the early injection stage, advection dominates oil displacement and reduces the oil molar fraction by approximately 24%; in the later stage, diffusion further reduces this fraction by more than 35%. The two mechanisms together raise the recovery factor. Without diffusion, over 70% of the residual oil would remain trapped in dead-end pores and become unrecoverable, which identifies diffusion as the key mechanism for enhancing CO_2-flooding recovery [22]. Sun et al. (2023) carried out high-pressure PVT experiments to examine how engineered SiO_2 nanoparticles, used as displacement-enhancing agents, strengthen the intrinsic interaction between CO_2 and crude oil. The nanomaterials are SiO_2 particles with sizes of roughly 20–30 nm; in field practice they are not injected alone but are co-injected with CO_2, usually as a nanofluid (e.g., dispersed in water) that is mixed with CO_2 before being pumped into the reservoir. Brownian motion induced by the nanoparticles generates local microconvection that disrupts the concentration boundary layer at the oil-phase interface, accelerating rather than changing the intrinsic diffusion and dissolution of CO_2 in the crude oil. Adding 0.1 wt% SiO_2 nanoparticles raised CO_2 solubility by 17% (enhancement coefficient for solubility, ES = 1.17), showing that the physical perturbation introduced by the nanomaterials improves CO_2–oil miscibility, increases crude-oil mobility, and enlarges the sweep volume, offering a practical route to a higher CO_2-flooding recovery factor [23]. Wang et al. (2023) used molecular dynamics simulations to examine viscosity variation in the CO_2 + 24-hexadecane system at 423.15–473.15 K and 1–5 MPa. As the CO_2 mole fraction rose from 0.07 to 0.32, the system viscosity decreased markedly. The system viscosity declined significantly when

the CO_2 mole fraction increased from 0.07 to 0.32. Taking the temperature of 423.15 K as an example, elevating the pressure from 1.23 MPa to 5.17 MPa reduced the viscosity from 1.03 mPa·s to 0.81 mPa·s, corresponding to a viscosity reduction of approximately 21.4%. CO_2 dissolution can thus alter the intermolecular potential energy of crude-oil molecules and reduce intramolecular friction, lowering the viscosity of the system [24]. Zhao et al. (2024) found experimentally that CO_2 dissolution widens the molecular spacing in crude oil and reduces intermolecular friction, causing a pronounced drop in the apparent viscosity of n-octane (39.05% in the bulk phase and 38.11% at the oil–water interface). It also raises the oil–water contact angle from 23.58° to 28.43°, lowers friction at the contact interface, reduces local flow resistance, and promotes crude-oil flow [25] (Fig. 2).

While these investigations have explored CO_2 dissolution behaviors in crude oil and the mechanisms whereby dissolution modulates fluid physical properties via macroscopic experiments and microscopic simulations alike, notable research gaps still exist. Most existing research addresses only individual CO_2 dissolution effects (viscosity reduction, swelling, diffusion) but seldom quantifies or compares the coupled processes of dissolution, swelling, extraction, diffusion, and viscosity reduction. The transition conditions between these effects remain unclear. Most experiments use idealized conditions and cannot represent the heterogeneity and pore-structure constraints of real reservoirs. Molecular dynamics simulations can reveal the microscopic origins of viscosity reduction, but effective methods for cross-scale quantitative mapping of these mechanisms to macroscopic recovery rates remain lacking. Addressing these gaps requires clarifying the transition conditions among CO_2 effects, strengthening experimental validation under real reservoir conditions, and developing cross-scale quantitative mapping methods that link microscopic mechanisms to macroscopic recovery—steps that would help translate laboratory findings into field engineering practice.

2.1.2 Swelling-induced viscosity reduction mechanism

CO_2 flooding is one of the physicochemical mechanisms used in enhanced oil recovery, drawing on the effects of CO_2 dissolution—specifically, crude-oil volumetric expansion and viscosity reduction. At the molecular level, small CO_2 molecules intercalate between hydrocarbon macromolecules, progressively replacing the original hydrocarbon–hydrocarbon van der Waals interactions with weaker hydrocarbon–CO_2 intermolecular forces; this process also widens the molecular spacing and induces volumetric expansion of the crude oil [26, 27]. The expansion directly lowers the internal frictional resistance encountered when hydrocarbon macromolecules move relative to one another, which appears macroscopically as a reduction in crude-oil viscosity [26, 28].

Zhang et al. (2013) used molecular simulation to calculate CO_2 solubility in crude oil and its effect on hydrocarbon expansion; drawing on phase-equilibrium theory and a

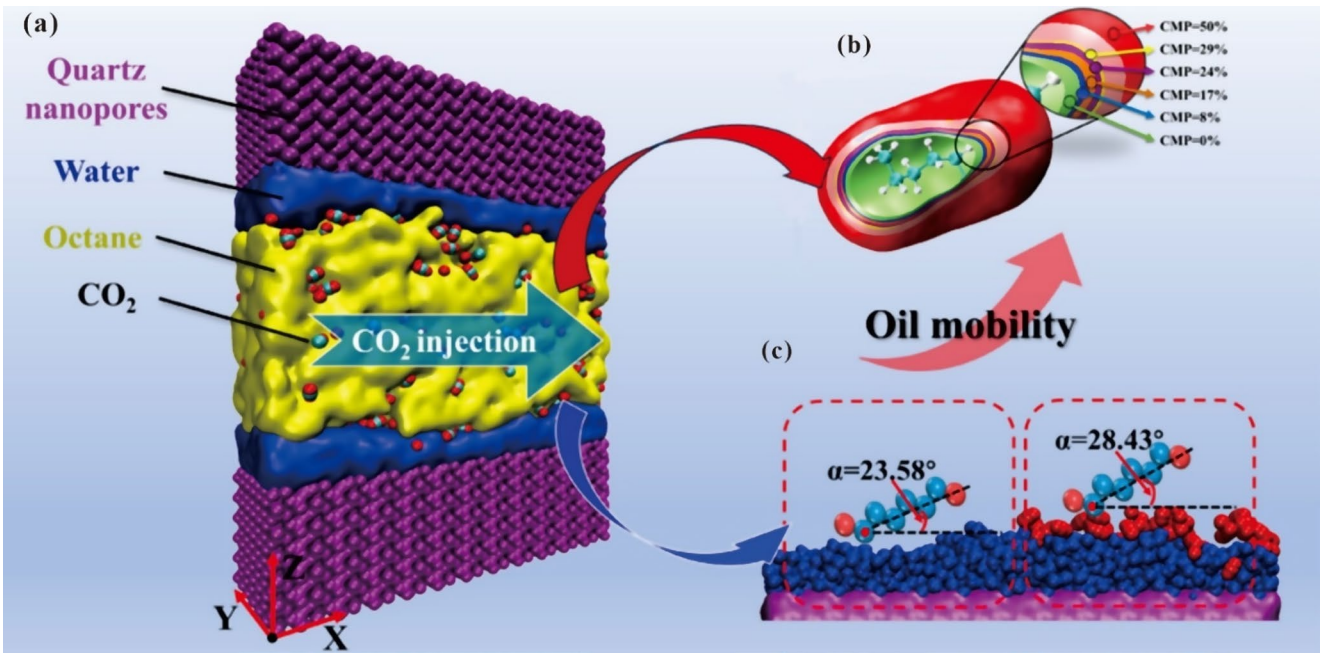


Fig. 2 Molecular Distribution and Velocity Profiles of nC8,Water, and CO_2 in Quartz Nanopores, (**a**) Mass Density Distributions of nC8, Water, and CO_2 in Quartz Nanopores, (**b**) Spatial distribution functions (SDFs) of nC8 for different CMP cases in equilibrium, (**c**) Orientation angle variation of octane molecules at the water film interface in quartz nanopores [25]

molecular diffusion model, they computed CO_2 solubility in octane and the resulting hydrocarbon expansion. At 323 K and 2–10 MPa, CO_2 solubility rose from 11.13 to 86.49 g/100 g, while the expansion factor increased linearly from 1.09 to 1.73. At the microscale, CO_2 dissolution in crude oil causes pronounced expansion of the oil phase; this expansion improves oil-phase mobility, lowers residual-oil retention in nanopores, and raises oil displacement efficiency [29]. Li et al. (2020) combined displacement experiments with computational simulations to examine CO_2–crude-oil miscibility and its effect on displacement efficiency. Integrating the volumetric expansion coefficient, diffusion coefficient, and minimum miscibility pressure, they identified the microscale miscibility mechanism: mutual solubility of CO_2 and crude oil raises the crude-oil volumetric expansion coefficient, lowers viscosity, and reduces the minimum miscibility pressure, which improves crude-oil mobility [30]. Wang et al. (2022) used molecular dynamics simulations to show that pore structure regulates the microscopic migration of CO_2 and the oil phase. Pore-surface wettability changes the migration pathways of CO_2 and the oil phase, whereas a semi-closed pore structure suppresses rapid CO_2 diffusion and strengthens oil-phase expansion, improving microscopic crude-oil mobility and expansion. These findings provide microscale mechanistic support for enhancing recovery via CO_2 flooding [31].Yuan et al. (2023) employed molecular dynamics simulations to examine the distribution and flow behavior of the CO_2–n-octane system in a 6-nm SiO_2 nanopore at 353 K. After CO_2 injection, CO_2 preferentially adsorbs onto the SiO_2 surface, desorbs adsorbed n-octane, and, through swelling, pushes the n-octane toward the pore center, where it mixes uniformly with bulk CO_2. The combined effect of swelling and viscosity reduction raises the flow velocity at the pore center; over an oil saturation range of 70%–100%, the n-octane mass flux increases continuously with CO_2 injection volume, confirming that CO_2-induced swelling and viscosity reduction enhance crude-oil mobilization and recovery in nanopores [32]. Wang et al. (2024) found experimentally that the crude-oil expansion factor in nanopores rises as pore radius decreases. At a CO_2 mole fraction of about 40%, the bulk expansion factor is roughly 1.08; in 10-nm pores it reaches about 1.28, some 18.5% higher than in the bulk phase. The expansion factor for shale oil, which is rich in light components, is about 1.28, markedly higher than the value of about 1.16 for heavy-component oils, an increase of roughly 10.3%. Within nanopores, CO_2 both expels residual oil through volumetric expansion and improves crude-oil mobility through dissolution-induced viscosity reduction; acting together, these two effects raise shale-oil recovery [33]. Zhou et al. (2025) used molecular dynamics simulations to show that CO_2 is miscible with crude oil and causes volume expansion, but sweep efficiency during single-phase CO_2 flooding is limited, leaving the crude oil in the pores difficult to mobilize fully. During CO_2–water alternating flooding, by contrast, CO_2 first weakens the interaction between crude oil and the pore walls, after which the advancing water phase carries the CO_2–oil mixture; this synergistic effect raises oil recovery efficiency. Under this scheme the oil recovery factor increases most sharply, confirming that alternating injection offers clear advantages for improving recovery in CO_2 flooding [34].

Existing research has elucidated the core mechanisms of enhanced oil recovery, including CO_2 dissolution-induced viscosity reduction and volume expansion, and has confirmed how injection methods and pressure conditions affect displacement efficiency; several gaps remain, however. First, most studies examine individual microscopic mechanisms in isolation while inadequately exploring the coupled control mechanisms involving pore structure, interfacial effects, and fluid–solid interactions; how these multiple factors interact synergistically or impose constraints on one another has yet to be fully resolved. Second, current research emphasizes the favorable aspects of CO_2 swelling-induced viscosity reduction but gives insufficient attention to its adverse effects—such as the formation of an interfacial layer through adsorption on pore walls, which reduces effective permeability. Future research should quantitatively characterize these multi-factor coupled control mechanisms and the competitive relationship between adsorption and viscosity reduction, and develop cross-scale simulation and experimental verification frameworks to support accurate prediction of CO_2 flooding performance and rational optimization of injection strategies.

2.2 Extraction miscibility mechanism

2.2.1 Extraction mechanism

In the multi-contact miscible CO_2 flooding process, extraction is one of the key microscale mechanisms that drive the convergence of gas- and liquid-phase compositions. Supercritical CO_2, whose density approaches that of a liquid and whose gas diffusion coefficient remains high, selectively extracts light and intermediate hydrocarbon components from crude oil and carries them into the gas phase, leaving the residual oil relatively enriched in heavy components. The extent of extraction determines whether a miscible zone can form and propagate; this process is controlled by the coupled action of several factors, including pressure, crude-oil composition, and pore-wettability characteristics.

Li et al. (2006) carried out a microscale experimental study of high-pressure CO_2 extraction of crude oil during supercritical CO_2 miscible flooding. At 20 MPa, CO_2

extracts more than 60% of the light components below C_{20} from the crude oil, improving its mobility; at 25 MPa the extraction capacity of CO_2 increases further, allowing partial mobilization of heavy components. As pressure rises, the CO_2–oil interfacial tension decreases sharply, leading to near-complete displacement of residual oil. Raising injection pressure both facilitates light-component extraction and serves as a key means of improving recovery in CO_2 miscible flooding [35]. Wang (2017) examined the mechanism of supercritical CO_2 flooding using molecular dynamics simulations. CO_2 adsorbs onto the rock surface through hydrogen bonding, separates oil molecules, and preferentially extracts light components such as heptane and toluene; its extraction capacity, however, depends on the injection rate. As the injection rate rises from 1.5 m/s to 6 m/s, heptane recovery increases from 66.7% to 82.7%, while toluene recovery falls from 29.4% to 8.8%; at 10 m/s, toluene recovery rebounds to 41.2% and heptane recovery reaches 87.3%. Extraction lowers crude-oil viscosity but is itself affected by the injection rate [36]. Yang et al. (2018) used molecular dynamics simulations to examine extraction behavior under different porosity conditions. Supercritical CO_2 permeates porous media by molecular diffusion and preferentially extracts light components below C_{15} from crude oil, lowering crude-oil viscosity and improving mobility. At 15% porosity, extraction efficiency is highest; increasing the CO_2 injection volume not only enhances light-component recovery but also allows simultaneous geological sequestration [37]. Li et al. (2023) performed microvisualization experiments to examine CO_2–crude-oil interaction at 25–75 °C and 1–30 MPa. CO_2 extracts light components from crude oil: as pressure increases, the gas phase changes from transparent to pale yellow, the oil phase lightens in color, the gas–liquid interface shifts from a hemispherical shape to a flat disk-like form, the contact angle rises from 37° (25 °C, 1 MPa) to 77° (25 °C, 30 MPa), and interfacial tension decreases markedly. CO_2 also desorbs oil molecules adsorbed on rock surfaces, improving crude-oil flowability [38]. Yang et al. (2024) used molecular dynamics simulations to examine how pressure affects CO_2 extraction capacity and its contribution to enhanced oil recovery, testing four pressures: 10, 20, 30, and 40 MPa. CO_2 extraction efficiency rises with pressure, but the gain from further pressure increases above 30 MPa gradually diminishes. At 20 MPa, the diffusion of crude-oil molecules is most favorable, which assists CO_2 mass transfer within the pores. At high pressure, CO_2 more readily displaces alkane molecules adsorbed on the rock surface, weakening adsorption between crude oil and the mineral surface, lowering residual oil saturation, and improving microscopic oil displacement efficiency [39]. Liu et al. (2025) used molecular dynamics simulations to examine the microscopic mechanisms of crude-oil-component extraction by supercritical CO_2. The distribution of various components of crude oil and the location of carbon dioxide were investigated at three time points (0, 1, and 2 ns) during the simulated extraction of crude oil using hydroxylated silica and methylated silica as wall surfaces for carbon dioxide. Supercritical CO_2 extracts light components from crude oil, lowers residual-oil viscosity, improves mobility in the pore space, and promotes desorption of oil molecules, raising microscale displacement efficiency [40] (Fig. 3). Tan et al. (2025) used molecular dynamics simulations to examine, at the molecular scale, the microextraction mechanism between CO_2 and crude oil in fractured low-permeability

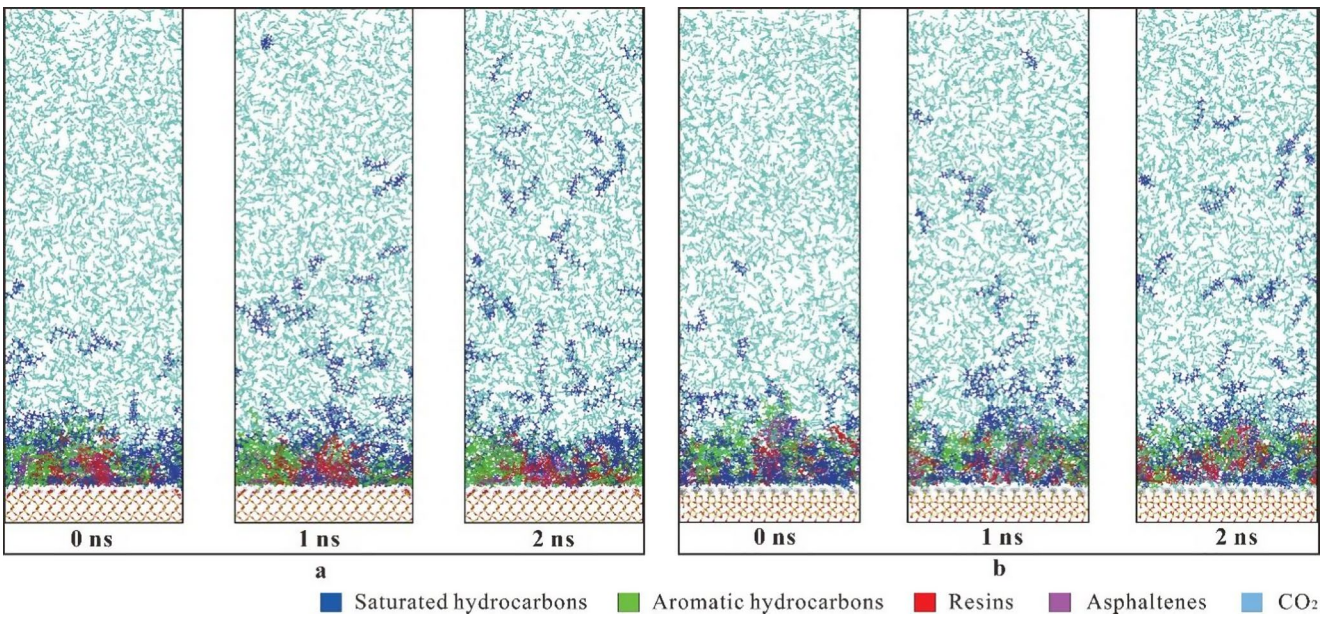


Fig. 3 (**a**) CO_2 extraction of hydroxylated SiO_2 surfaces crude oil; (**b**) CO_2 extraction of methylated SiO_2 surfaces crude oil [40]

reservoirs. CO_2 extracts light components from crude oil, lowers oil–gas interfacial tension, weakens adsorption of the oil phase on the rock surface, and reduces flow resistance, increasing both microscale displacement efficiency and macroscopic sweep volume. These results provide molecular-level support for enhanced oil recovery via CO_2 flooding [41].

At the molecular scale, studies have confirmed that supercritical CO_2 enhances oil displacement efficiency by extracting light components and stripping oil molecules from pore surfaces; several research gaps nonetheless remain. First, most existing studies examine the forward displacement mechanism of CO_2 while neglecting adverse effects induced by light-component extraction—specifically, asphaltene precipitation—and provide little insight into how such precipitation might be contained or suppressed. Second, current studies describe the behavior of individual parameters such as pressure and porosity but do not define the optimal operating range under multi-parameter coupling; the paradoxical observation that displacement enhancement slows at high pressure while low-pressure diffusion yields the best results also remains unexplained. Third, because existing work relies on conventional reservoir simulations and experiments, the displacement adaptability of fractured and complex-porosity reservoirs remains poorly characterized, which hinders the precise application of these mechanisms to diverse field development conditions. Future research should prioritize asphaltene-precipitation inhibition, the identification of optimal multi-parameter coupling conditions, and an explanation of the high-pressure/low-pressure contradiction, while extending adaptability studies to fractured reservoirs so that microscopic displacement processes can be more closely linked to actual field operating conditions.

2.2.2 Emulsification mechanism

In CO_2 flooding, miscibility is one of the mechanisms that markedly increase oil recovery. When injected CO_2 mixes with reservoir crude oil, the interfacial tension between the two phases approaches zero, capillary forces are eliminated, and a single fluid phase with uniform physical properties ultimately forms. Miscibility, however, is not simply a fluid-mixing process but a complex phenomenon involving multi-component mass transfer and phase evolution; the key to achieving miscibility is the minimum miscibility pressure (MMP), a parameter that directly determines the compatibility of reservoir conditions with the injection process.

Chen et al. (2022) performed core displacement and slim-tube experiments. Based on the multi-contact miscibility mechanism, the CO_2–crude-oil contact zone was divided into four dynamic phase zones. Although near-miscible flooding reached a final recovery factor of 74.26%, slightly below that of miscible flooding (86.50%), its capillary flow capacity and overall recovery factor were markedly higher than those of non-miscible flooding [42]. Xue et al. (2023) conducted molecular dynamics (MD) simulations to investigate CO_2 miscible displacement in calcite nano-slits. Their simulation results reveal that CO_2 first displaces octane adsorbed on pore surfaces, followed by gradual diffusion and mixing with crude oil until full miscibility is achieved. Van der Waals interactions between the two phases dominate the miscible mixing process. Increasing temperature and slit aperture both accelerate miscibility evolution, while an optimal miscible pressure exists for the system. Collectively, these findings demonstrate that carbonate nano-pores provide favorable microscale conditions for CO_2 miscible flooding (Fig. 4)[43]. Gandomkar et al. (2024) performed miscible displacement experiments on carbonate rock cores. Under miscible conditions at 3,800 psi and 80 °C, CO_2-flooding recovery reached 88.7%, 26.5% points higher than pure CO_2 flooding (62.2%). The improvement arises because PDMS lowers the oil–gas interfacial tension nearly to zero and reduces crude-oil viscosity from 3.8 cp to 3.0 cp, raising recovery from 36% to 50% and improving sweep efficiency. At 3,250 psi—still above the minimum miscibility pressure (MMP)—however, CO_2-flooding recovery fell to 66.9%, showing that higher pressure favors molecular miscibility and a higher degree of mixing between CO_2 and crude oil, and therefore a higher overall recovery [44]. Torabi et al. (2024) examined how four small-molecule CO_2-soluble polymers—PDMS, PFA, P-1-D, and PVEE—enhance CO_2 miscible flooding. These polymers dissolve stably in CO_2 under gas-flooding conditions, lowering the CO_2–oil interfacial tension and reducing the minimum miscibility pressure (MMP) by 10%–19.4%. The CO_2/PDMS system raised recovery by 26.5% points relative to pure CO_2 flooding under miscible displacement, indicating that small-molecule CO_2-soluble polymers with oxygen-containing functional groups lower the miscibility threshold and improve oil displacement efficiency [45]. Zhao et al. (2025) examined the transition of CO_2 flooding from miscible to non-miscible regimes as formation pressure declines; using the reduction in minimum miscibility pressure (MMP) as the key indicator, they assessed how associated-gas composition and chemical additives regulate miscibility conditions. Reinjecting de-methanized associated gas (C_2–C_5) lowers the MMP by more than 9.97%; light hydrocarbons dissolve and enrich the CO_2 mixture, strengthening extraction and reducing the pressure required for miscibility. A composite system of 0.2% ethylene glycol butyl ether + 0.4% ethanol reduces the MMP by 17.38%, as the two additives act together to lower interfacial tension and weaken the phase-interface energy barrier (the barrier imposed by interfacial

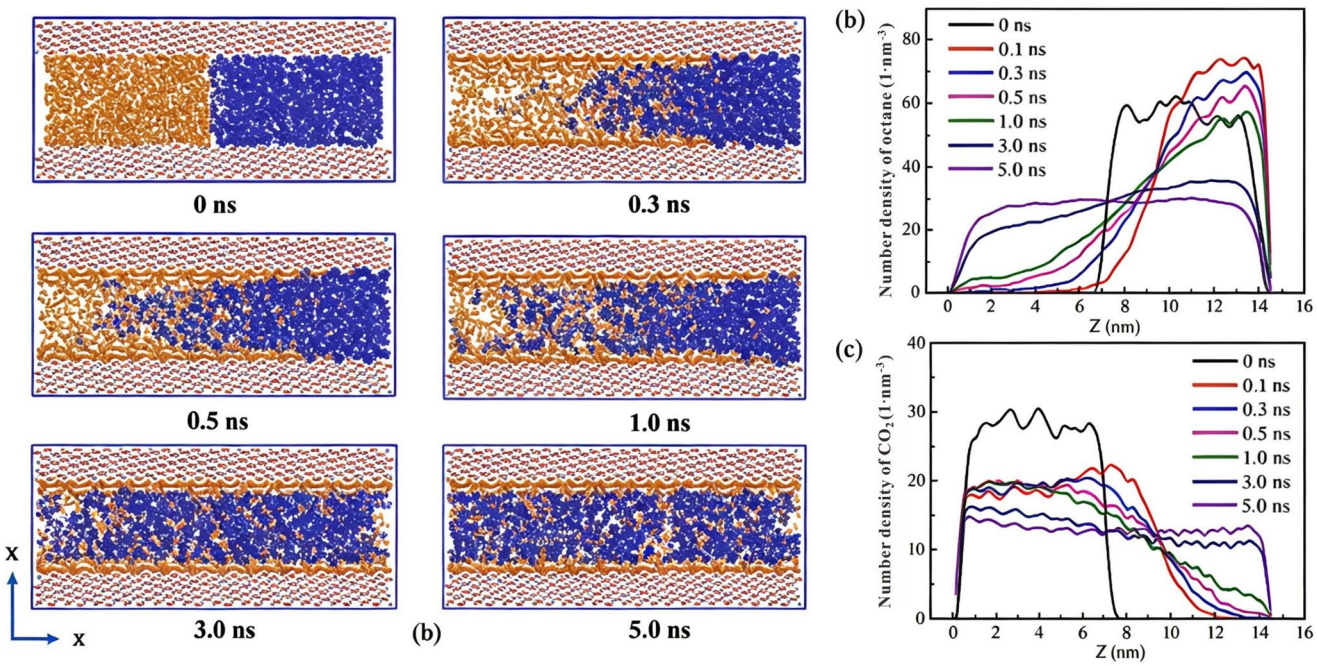


Fig. 4 (**a**) The snapshot of the miscible process of CO_2 and octane, the number density profiles along the Z direction. (**b**) Octane, (**c**) CO_2 [43]

tension), facilitating miscibility [46]. Wang et al. (2025) proposed a method for enhancing CO_2–tight-oil miscibility based on a direct-current electric field. The field converts heavy components (C_{26}–C_{31}) in crude oil into lighter components (C_{16}–C_{21}), lowering crude-oil viscosity by 16.1% and reducing the system minimum miscibility pressure (MMP) from 22.5 MPa to 21.6 MPa. The direct-current field strengthens the interaction energy between CO_2 and crude-oil molecules, raising the diffusion-coefficient slope of crude-oil molecules to 0.986 (a 65% increase). Improved molecular miscibility and diffusion at the microscale intensify oil-phase swelling, enlarge the free volume of the system, and lower the miscibility energy barrier at the oil–gas interface, enhancing the miscibility performance of the CO_2–crude-oil system at the microscale [47]. Jia et al. (2026) performed supercritical CO_2 foam miscible flooding at pressures above the minimum miscibility pressure. The apparent viscosity peaked at an optimal supercritical CO_2 foam mass fraction of about 0.75; oil-phase composition affected foam strength, and both foam apparent viscosity and recovery factor were higher during C_{16} displacement than during C_{10} displacement. In-situ generated foam gave higher recovery for both C_{10} and C_{16} (17.78% and 30.91%, respectively) than pre-generated foam (15.91% and 20.83%, respectively); the liquid film of the pre-generated foam hindered direct contact between CO_2 and crude oil, weakening miscibility [48]. Wang (2026) showed experimentally that the miscible-flooding mechanism in CO_2 flooding is a dynamic miscibility process driven by coupled vaporizing-gas-drive and condensing-gas-drive mechanisms. Injected CO_2 continuously extracts light components from crude oil (vaporizing gas drive) while promoting the condensation of intermediate hydrocarbons from the CO_2-rich gas phase into the oil phase (condensing gas drive); through repeated contact and mass transfer, the compositions of the two phases converge, interfacial tension falls nearly to zero, capillary resistance is eliminated, and oil recovery reaches a very high theoretical level [49].

Although prior work has described the miscible-flooding mechanism of CO_2 injection in reasonable detail, several challenges remain when the process is applied under actual reservoir conditions. No quantitative method yet exists to characterize the trade-off between flow-resistance control and mass-transfer efficiency in foam-assisted miscible flooding. Cross-scale parameter links are also lacking between the microscopic mechanisms identified by molecular simulation and macroscopic fluid-flow behavior. Future work should develop a miscible-flooding evaluation framework that couples multiple scales and mechanisms, supporting the shift of CO_2 miscible flooding from laboratory and simulation studies to field application.

2.3 Competitive adsorption mechanism of CO_2 flooding for crude oil displacement

Competitive adsorption is central to CO_2 flooding in shale oil production and is a key microscale mechanism for releasing organic matter and displacing crude oil adsorbed on mineral surfaces. Shale reservoirs feature tiny pores and weak pore connectivity, making adsorbed hydrocarbons difficult

to recover. With a smaller molecular diameter, CO_2 exhibits stronger adsorption affinity for kerogen and clay surfaces, enabling it to preferentially displace hydrocarbons into the free phase. CO_2 is also adsorbed by the pore walls, producing a combined effect of displacement-enhanced production and adsorption-based carbon sequestration that raises shale-oil recovery and stores carbon [50].

Gandomkar et al. (2013) used cyclic adsorption experiments to examine the electrostatically dominated adsorption of an anionic surfactant (sodium dodecyl sulfate) on carbonate rock surfaces. In surfactant-assisted CO_2 flooding, adsorption losses at the rock surface can exceed 90% of the total injected volume; excessive adsorption raises operational costs, alters wettability, and weakens interfacial control, lowering both CO_2 sweep efficiency and microscale oil displacement, which limits recovery improvement and undermines sequestration safety. Adsorption losses must therefore be included as a key factor when evaluating chemical-enhancement potential; they can be reduced by sacrificial agents or pre-adsorption, and by optimizing injection concentration and slug design, to improve both oil displacement efficiency and sequestration reliability [51]. Liu et al. (2016) used molecular dynamics simulations to examine the effects of supercritical CO_2 on the water–n-octane interface. CO_2 preferentially adsorbs and enriches at the interface, forming an interfacial adsorption layer; as temperature rises, the CO_2 mole fraction at the interface increases from 0 to 0.8, the interface width widens from 4.02 Å to 11.73 Å, the orientation angle of n-octane molecules decreases from 83.25° to 27.75°, and interfacial tension falls linearly (from 50.88 mN/m for pure water–n-octane to 32.58 mN/m for water–CO_2). By lowering interfacial tension and enhancing diffusion, CO_2 promotes the initiation and migration of residual oil, raising oil recovery [52]. Zhao et al. (2021) built three molecular models for kerogen adsorption and hydrocarbon extraction using the COMPASS II force field, performed molecular dynamics simulations of reverse diffusion of CO_2 and hydrocarbons in shale kerogen, and examined the microscopic driving mechanisms of this diffusion in terms of intermolecular interaction energy. The kerogen–CO_2 interaction is stronger than the kerogen–hydrocarbon interaction, which is the core mechanism behind the spontaneous reverse diffusion of CO_2 and hydrocarbons in kerogen; this process allows adsorbed hydrocarbons to be extracted from kerogen while CO_2 is stored in the kerogen in the adsorbed phase, producing a microscale synergistic effect of oil displacement and carbon sequestration [53] (Fig. 5).

Zhang et al. (2021) used non-equilibrium molecular dynamics (NEMD) simulations with quartz nanochannels to explore micro-adsorption and oil-displacement mechanisms under different CO_2 injection rates. CO_2 adsorption capacity in nanopores decreases at higher injection rates, which weakens CO_2-crude-oil interfacial interactions, yet strengthens macroscopic driving force to facilitate heavy-oil transport and enhance recovery [54]. Zhang et al. (2021) used molecular dynamics simulations to examine the microscale behavior of the CO_2–octane two-phase system in a 5 nm calcite nanopore, identifying the competitive-adsorption patterns of CO_2 and octane under nanoconfinement. Under fluid–structure interaction, the fluid near the pore wall shows distinct density stratification; CO_2 molecules, with their stronger adsorption affinity, preferentially occupy adsorption sites on the pore wall, displacing adsorbed octane molecules and promoting detachment of the oil phase from the wall toward the pore center [55]. Wang et al. (2022) adopted molecular-dynamics simulations to study CO_2-driven crude-oil displacement in nanopores. CO_2 diffuses far faster than alkanes, with its rate dependent on pore size: the 0.9 nm pore yields optimal displacement via stratified flow, while mutual obstruction in the 0.65 nm single-layer-adsorption pore leads to the slowest displacement. Owing to stronger wall-CO_2 interaction than wall-crude-oil interaction, CO_2 competitively displaces adsorbed oil films and pushes crude oil toward pore centers [56]. Du et al. (2022) used molecular dynamics simulations to examine the microscale competitive-adsorption mechanism of CO_2 in organic nanopores for shale-oil displacement. CO_2 adsorption on pore surfaces is much stronger than that of crude-oil components, and displacement efficiency rises with pressure and pore diameter. CO_2 adsorption first increases and then decreases with temperature, peaking at 393 K, which reflects the complex effect of temperature on the adsorption–displacement equilibrium [57]. Dong et al. (2023) used non-equilibrium molecular dynamics simulations to build models of kerogen-based organic pores and four types of inorganic nanopores, examining how nanopore type and displacement velocity affect the microscale mechanisms of CO_2-driven shale-oil displacement. Inorganic nanopores generally outperform kerogen-based organic pores in displacement efficiency, with feldspar pores performing best, whereas calcite pores are more prone to CO_2 channeling. CO_2 also displaces light hydrocarbon components more effectively than heavy components; lower displacement velocities extend the contact time between CO_2 and crude oil, strengthening competitive adsorption and displacement and raising overall displacement efficiency [58] (Fig. 6).

Huang et al. (2023) used molecular dynamics simulations to examine the adsorption of crude-oil components in nanopores of organic-rich shale. The shale pore walls exert strong fluid–solid interactions that promote a solidified interfacial oil layer on the wall surface, which acts as microscale resistance to crude-oil displacement. Upon CO_2 injection, CO_2 molecules possess superior adsorption affinity and gradually replace oil molecules within the adsorbed

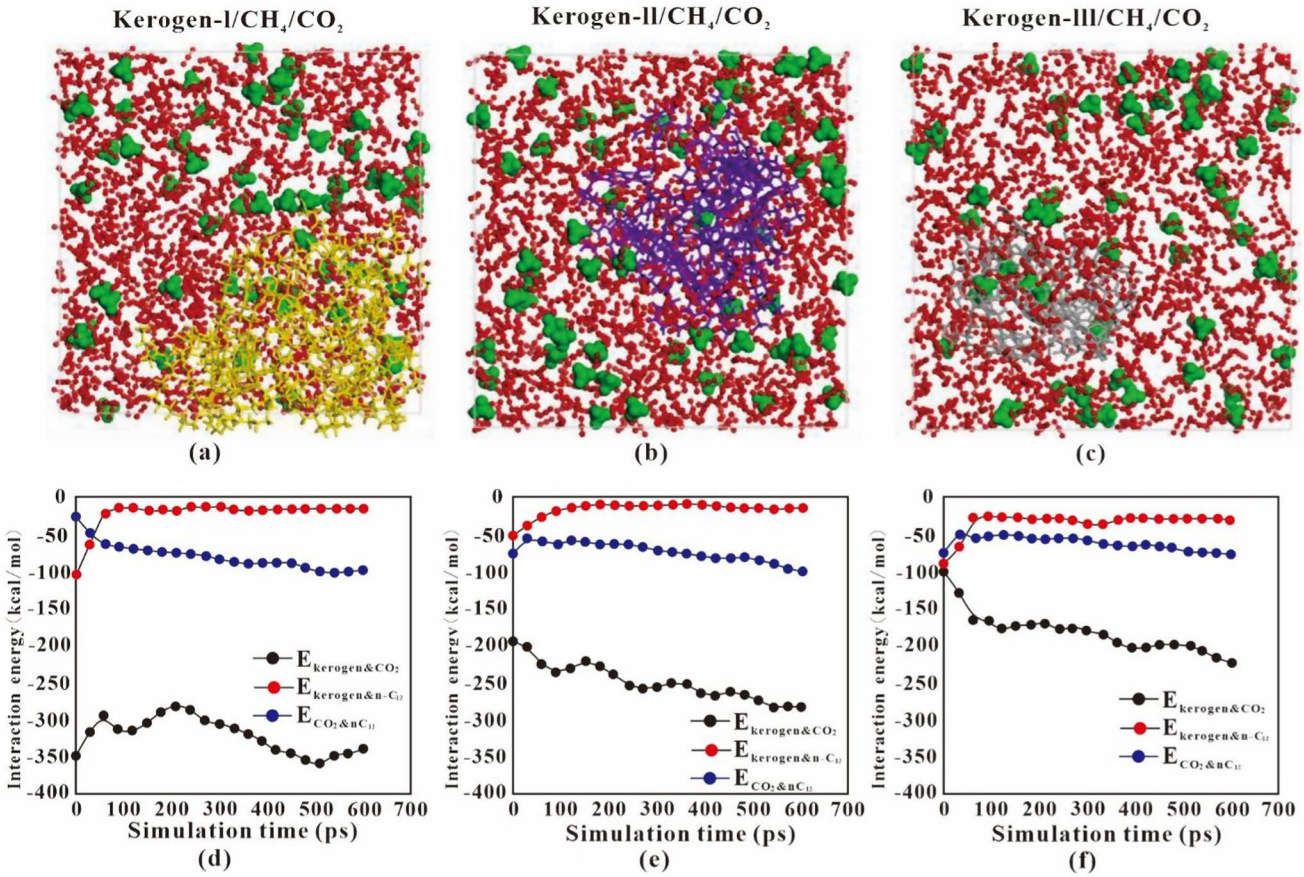


Fig. 5 Simulation results of CH_4 extraction from different types of kerogen using CO_2 at 20 MPa and 353.15 K, and corresponding interaction energies: (a-1) Equilibrium structure for kerogen-I/CH_4/CO_2; (a-2) Interaction energies for kerogen-I/CH_4/CO_2; (b-1) Equilibrium structure for kerogen-IIA/CH_4/CO_2; (b-2) Interaction energies for kerogen–IIA/CH_4/CO_2; (c-1) Equilibrium structure for kerogen-IIB/CH_4/CO_2; (c-2) Interaction energies for kerogen IIB/CH_4/CO_2. Different types of kerogen are depicted by different colors: yellow ─ atoms and bonds of kerogen-I; blue ─ atoms and bonds of kerogen-IIA; gray ─ atoms and bonds of kerogen-IIB [53]

layer via competitive adsorption. This causes crude oil to desorb from pore walls, migrate toward the central zone of pore throats, and reduce flow resistance inside nanopores [59]. Huang et al. (2024) used molecular dynamics simulations of nanopore systems with pore diameters of 2–20 nm to examine the microscale dynamic distribution of complex crude-oil components after CO_2 injection, focusing on changes in the thickness of the crude-oil adsorption layer. After CO_2 injection, the thickness of the crude-oil adsorption layer in the nanopores decreases gradually as pore diameter increases; larger pores weaken adsorption of crude oil by the rock wall, strengthening microscale displacement between CO_2 and crude oil, thinning the adsorption layer, and raising the proportion of free oil phase [60]. Sakthivel et al. (2024) used a high-temperature, high-pressure goniometer and an interfacial-tension apparatus to examine how carbon nanodots (CNDs) affect the contact angle in carbonate rock–CO_2–oil systems. During CO_2 flooding, crude-oil components adsorb strongly onto the carbonate rock surface, forming a dense oil film that produces strong CO_2-wetting behavior at high pressure (contact angle up to 122°) and severely hinders displacement. CNDs counter this by adsorbing competitively and stripping the oil film from the rock surface, lowering the contact angle to 86° at 1000 ppm and 3000 psi and shifting wettability from oil-wet to weakly water-wet. The high capillary pressure generated by hydrophilic pores has a dual effect: during flooding it impedes CO_2 entry into nanopores, reducing residual-oil recovery, whereas during sequestration it can form a CO_2-impermeable barrier. As shown in Fig. 7, the SEM images in Fig. 7(a) and 7(c) reveal abundant nanoscale pore throats in the carbonate reservoir; the curves in Fig. 7(d) and 7(f) show that water contact angles for both pure carbonate and CND-modified rock samples stay below 45°, indicating that the hydrophilic state is little affected by changes in temperature and pressure. These pore throats generate sustained reverse capillary resistance that opposes the upward migration of CO_2 as formation pressure declines, allowing bound-water mechanisms to form microscale carbon seals. In crude-oil-aged wet rock samples, by contrast,

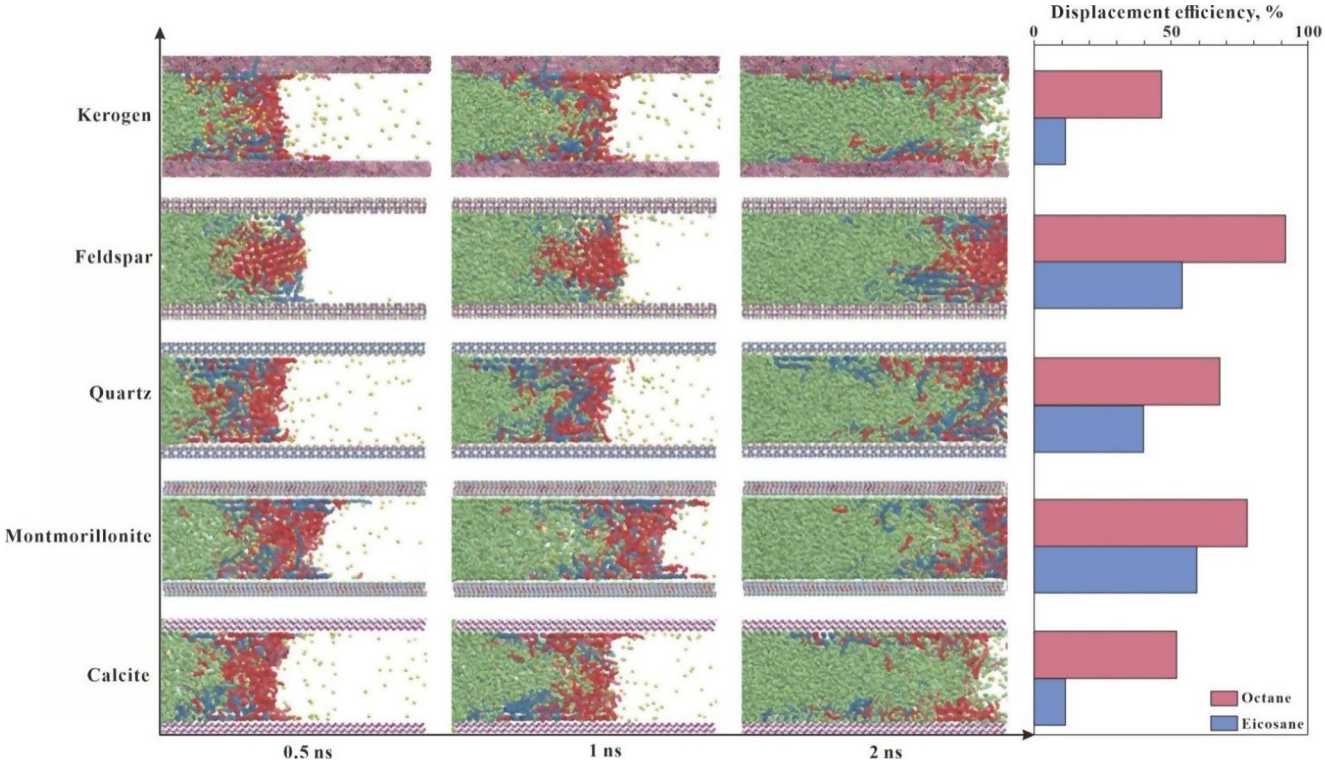


Fig. 6 Molecular dynamics configurations and displacement efficiency of octane and eicosane during CO_2 flooding in different shale nanopores at 0.5 ns, 1 ns, and 2 ns. The color markings are CO_2 (green), methane (yellow), octane (red), and eicosane (dark blue) [58]

the contact angle remains above 90° (Fig. 7(b) and 7(e)); under these conditions capillary force promotes CO_2 seepage, so no effective seal forms and the leakage risk rises. Hydrophilic pores and the high capillary pressure produced by the CND-modified structure are thus the key microscale conditions for long-term CO_2 storage [61] (Fig. 7). Zhang et al. (2024) used molecular dynamics simulations to examine competitive adsorption and transport of nC_{10} alkane molecules relative to CO_2 in nanopores of 0.7–5 nm. Pore size strongly modulates the intensity of this competitive adsorption: as pore size decreases, nC_{10} adsorption capacity first rises and then falls; at 1 nm, CO_2 adsorption capacity increases sharply and competitive adsorption is strongest, so CO_2 most effectively displaces oil molecules adsorbed on the pore walls [62]. Huang et al. (2024) used molecular dynamics simulations to examine the dynamic distribution of complex crude-oil components in shale nanopores during CO_2 flooding, tracking microscale mass transfer and interaction between CO_2 and the oil phase. After CO_2 injection, CO_2 uses its competitive-adsorption advantage to displace and desorb crude oil adsorbed on the shale pore walls, thinning the oil layer adsorbed on the wall surface; this releases the originally adsorbed oil phase and converts it into mobile bulk fluid, achieving the core goal of desorption-induced production enhancement. During the depressurization production stage, however, residual components readsorb onto the pore walls once the competitive inhibition by CO_2 is lost, forming new adsorption layers that reduce the proportion of mobile oil in the total volume and make further recovery gains difficult [60] (Fig. 8).

Experimental and molecular-simulation studies have clarified the microscopic CO_2 competitive-adsorption mechanism for oil displacement, but several limitations remain. The competitive-adsorption advantage is strongly condition-dependent, governed by temperature, pore size, and mineral composition, rather than being a universal rule, and existing work still lacks a systematic characterization of its boundary conditions. Most current studies focus on forward competitive adsorption by CO_2 during injection and overlook the recombination–decomposition lag and the readsorption risk that appear during depressurization production as CO_2 concentration falls, so the effect of these phenomena on long-term recovery needs further evaluation. Future work should move from single-medium static characterization toward dynamic system-level assessment that accounts for multiple interacting factors, building an integrated research framework for oil displacement and carbon sequestration.

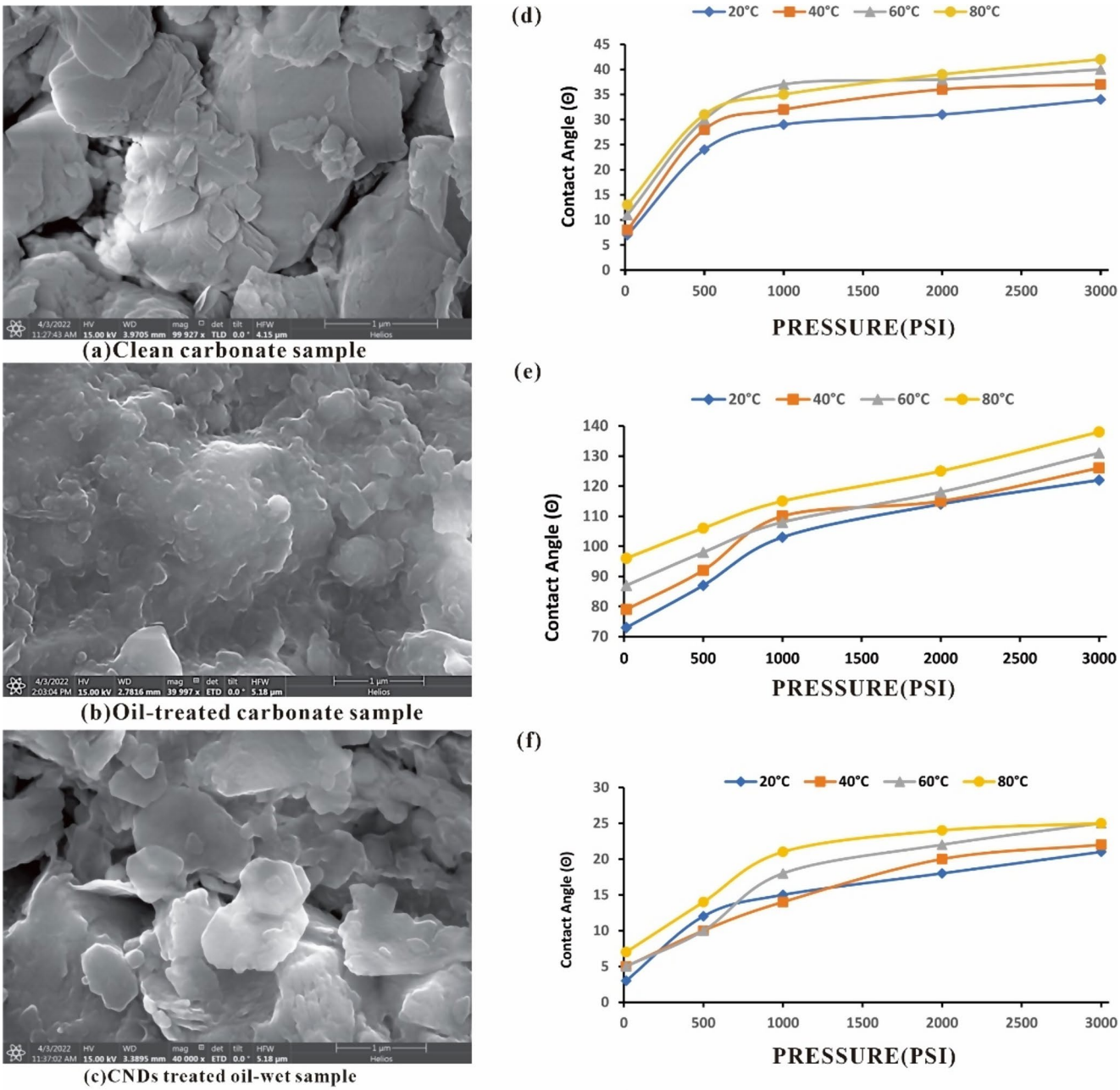


Fig. 7 Carbonate/CO_2/brine wettability and CO_2–brine interfacial tension modification by carbon nanodots: (**a**) Clean Carbonate Wettability; (**b**) Oil-wet Carbonate Wettability; (**c**) CNDs-treated Clean Carbonate Wettability; (**d**) CNDs-treated Oil-wet Carbonate Wettability; (**e**) Effect of CNDs Concentration on Wettability; (**f**) Effect of CNDs on CO_2–Seawater Interfacial Tension [61]

2.4 Mass transfer mechanism of CO_2 in nanopores

Shale oil reservoirs contain abundant micro- and nanopore throats, so fluid migration in these confined spaces differs markedly from that in conventional oil reservoirs. Supercritical CO_2 has strong diffusion and dissolution capacity, allowing it to invade nanopores and drive shale-oil displacement. This mobilization follows a distinct mass-transfer pathway from fractures to micropores: CO_2 migrates along the fractures and then diffuses into the nanopores, where confinement induces a nano-confinement effect that helps CO_2 reach the supercritical state, strengthening its dissolution capacity and lowering interfacial tension. With its strong adsorption affinity for pore walls, CO_2 also separates oil molecules from the wall surfaces and uses the combined effect of permeation–dissolution–separation to mobilize crude oil in the nanopores.

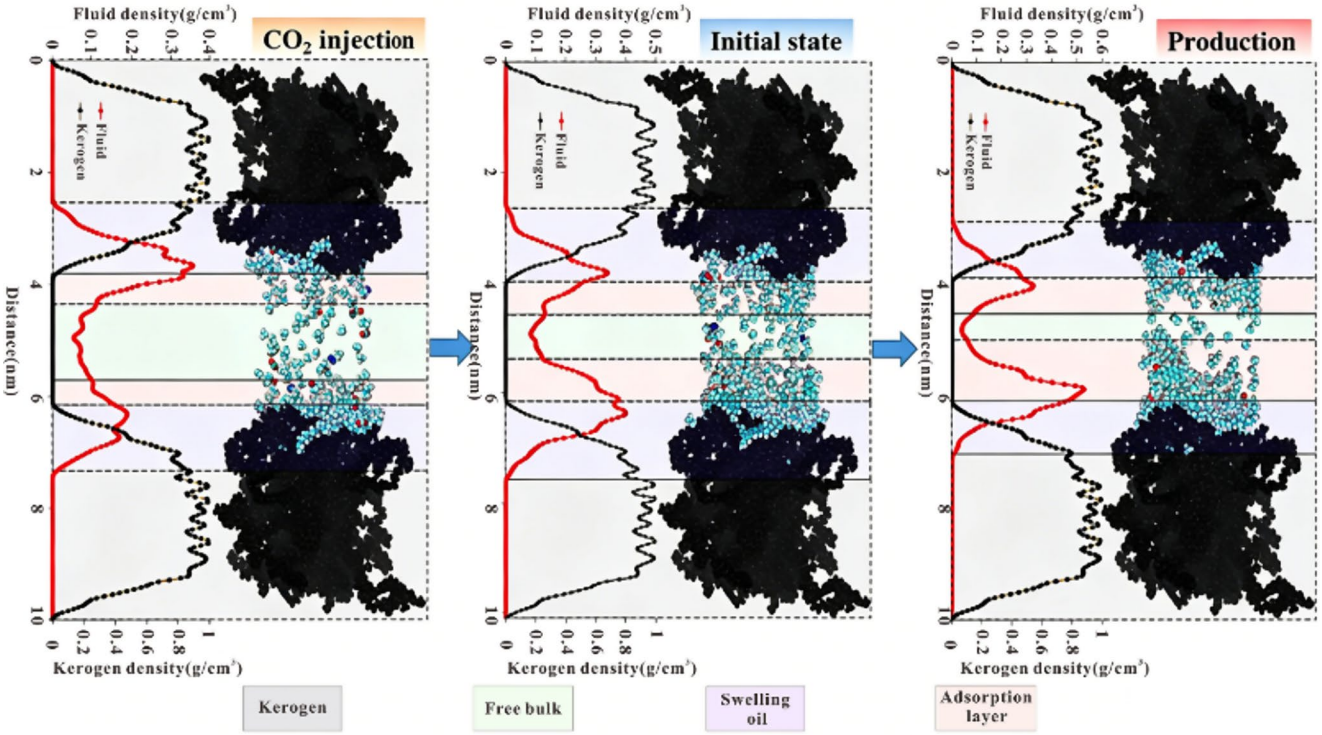


Fig. 8 The adsorption characteristics of shale oil in kerogen at three production stages (2 nm). Components color scheme: black, kerogen I-A. Elements color scheme: cyan, C; blue, N; red, O; white, H; purple, K; pink, Al; orange, Si (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article) [60]

Wang et al. (2022) used molecular dynamics simulations to build quartz nanopore models of 0.65–15 nm and examined the microstructure of n-decane under nanoconfinement and its effect on CO_2 displacement. As pore size increased, n-decane molecules shifted from highly ordered monolayer/double-layer arrangements to a disordered distribution, while the molecular number density and diffusion coefficient showed periodic oscillations. Under these structural effects, CO_2 displacement efficiency is inversely related to pore size, reaching nearly 100% in micropores (< 2 nm). Raising displacement pressure also improves efficiency across all pore sizes, with the highest overall efficiency at 14 MPa [56]. Cai et al. (2025) combined fractal theory with multi-physics coupled numerical simulations to characterize micro- and nanopore-scale mass transfer, including CO_2–crude-oil phase transition and molecular diffusion. Nanopore confinement markedly strengthens molecular diffusion; by raising residual-fluid density, CO_2 increases the CO_2 mole fraction in the displacement zone by up to 46.8%, improving crude-oil mobility and concentration-distribution uniformity and enlarging the sweep volume. This raises CO_2 displacement recovery at the micro-mass-transfer level and offers theoretical support for CO_2 flooding in tight reservoirs [63]. Liu et al. (2025) developed a KH550 (silane coupling agent)–PEI (polyethylenimine, an amine-rich polymer) double-modified SiO_2 nanofluid with a particle size of 15 nm. Amine groups on the nanoparticle surface react reversibly with CO_2 to form carbamate compounds, giving a high CO_2 loading capacity (up to 3.3 mmol/g) and deep, sustained release. The nanofluid delivers CO_2 to low-permeability zones, delays gas breakthrough, improves CO_2 utilization efficiency, enlarges the affected volume, and raises CO_2-flooding recovery [64]. Bi et al. (2026) used nuclear magnetic resonance computed tomography (NMR-CT) together with characterization techniques to examine the selective utilization and migration of CO_2 during miscible flooding in deep low-permeability cores at the pore scale. CO_2 preferentially displaced crude oil in large pores (> 90%); as permeability increased, the proportion of medium-sized pores being utilized rose (the average displacement ratio increased from 69.8% to 79.1%). Oil displacement efficiency rose with permeability (from 66.53% to 86.28%); in the low-permeability reservoir, however, high pore resistance left residual oil mainly in micro- and nanoscale pores, where differences in pore-scale flow dynamics limited CO_2 migration and miscible displacement [65]. Meng et al. (2026) performed microscale flooding experiments on low-permeability cores to examine the pore-scale displacement and migration of the supercritical CO_2–ADA gel system. In a heterogeneous reservoir with a permeability gradient of 10.6, first-stage CO_2 flooding gave 20.8% recovery; after injecting a mixture of 0.6 wt% ADA

polymer and 1.2 wt% phenolic resin gel, the polymer-based viscosity-increasing phase raised recovery by 4.6%. In the second stage, after gelation, CO_2 flooding increased recovery by a further 23.1%, for an ultimate recovery of 48.5%. When the permeability gradient rose to 15.2, ultimate recovery fell to 40.9% because of earlier gas breakthrough. Flow-path regulation by the gel improved CO_2-flooding recovery efficiency [66].

Research on CO_2-flooding recovery mechanisms—including microscale mass transfer, nanofluid-based regulation, and gel-based regulation—has shown that pore size and pressure strongly affect displacement efficiency, but several limitations remain. Studies at different scales proceed in isolation, with no cross-scale coupling framework linking molecular arrangement to macroscopic flow behavior, which prevents the theoretical chain of diffusion enhancement and sweep expansion from being integrated. Modified nanofluid and gel-based regulation techniques have also shown only individual enhancement effects, and their long-term stability and reservoir compatibility have not been examined systematically, limiting their use in guiding high-quality, high-efficiency CO_2 flooding in heterogeneous low-permeability reservoirs. Future work should move from static mechanistic analysis to multiscale dynamic characterization and reservoir-compatibility assessment, building a unified framework that links the micro- and macroscales.

3 Research status of CO_2–water microscopic reaction mechanisms

3.1 Transport behavior of CO_2 and formation water

In CO_2-enhanced oil recovery (CO_2-EOR), the migration of CO_2 and formation water in the restricted pores of the reservoir directly governs displacement efficiency. Under the coupled action of capillary forces and interfacial effects, two-phase flow patterns differ markedly from those in conventional reservoirs. Raising the CO_2 injection rate lowers the interfacial tension between reservoir fluids and formation water, while pressure variations further affect interfacial properties and the flow regime by changing the two-phase density difference. A clearer understanding of migration mechanisms and interfacial evolution in CO_2–formation-water systems under confinement offers useful theoretical support for optimizing CO_2 injection strategies and improving crude-oil recovery.

Niu et al. (2017) performed high-density supercritical CO_2 flooding and supercritical CO_2/water alternating injection experiments using a high-temperature, high-pressure microscopic visualization model. A vertically oriented laser-etched microscopic model was used to reproduce formation temperature and pressure, and thus the supercritical CO_2 state, while the distribution of CO_2 and water in pores and throats, the Jamin effect, and residual-oil displacement were observed. High-density supercritical CO_2 diffuses and migrates toward the lower sections of the reservoir, improving longitudinal sweep efficiency [67]. Cheng et al. (2017) combined nuclear magnetic resonance imaging with long-core flooding experiments. Nuclear magnetic resonance T_2 spectroscopy was used to monitor changes in pore-fluid distribution during CO_2 flooding and to examine the microscale interaction between CO_2 and residual oil in the presence of a water phase. CO_2 can penetrate the water film and contact isolated residual oil at pore and throat boundaries, causing dispersed oil droplets to coalesce and be displaced; CO_2 injection also induces precipitation of ions such as Ca^{2+}, Mg^{2+}, and Mn^{2+} from highly mineralized formation water, altering pore-fluid properties and the rock-surface environment at the microscale [68]. Luan et al. (2020) used molecular dynamics simulations to build a silica-based blind-end nanopore model with an aqueous film and to simulate supercritical CO_2 displacement of n-octane. Based on intermolecular interactions and hydrogen-bond evolution, the spatiotemporal distribution of CO_2, water, and oil molecules was tracked. Supercritical CO_2 first diffuses into the aqueous film, disrupts the hydrogen-bond network among water molecules, induces micropores within the film, and gradually connects them to form a continuous mass-transfer pathway (the water-film rupture process observed at 1700–1706 ps in Fig. 9). CO_2 also aggregates at the oil–water interface and changes interfacial tension, causing the oil phase to expand and be displaced; this is the microscopic mechanism by which CO_2 displaces residual oil efficiently through water-film rupture [69] (Fig. 9). Liu et al. (2022) used molecular dynamics simulations to build a quartz shale nanopore model with coexisting oil and water bridges and to simulate supercritical CO_2 injection and displacement. Based on intermolecular interactions, hydrogen-bond evolution, concentration gradients, and pressure-gradient driving, they analyzed how CO_2 overcomes the water bridge at the microscale: ① water molecules at the center of the water bridge show a porous distribution, with few hydrogen bonds and weak stability, which provides a pathway for CO_2 invasion; ② supercritical CO_2 disrupts the central structure of the water bridge through a dual mechanism of high-speed scouring and hydrogen-bond dragging, causing the bridge to fracture and a fluid channel to form; ③ CO_2 does not compete strongly with hydroxyl groups on the rock surface; after the water bridge breaks, dissociated water molecules adsorb onto the hydrophilic rock wall and thicken the water film, while CO_2 forms a gas cushion between the water film and the oil phase, lowering flow resistance; ④ under the combined action of pressure driving and mutual diffusion,

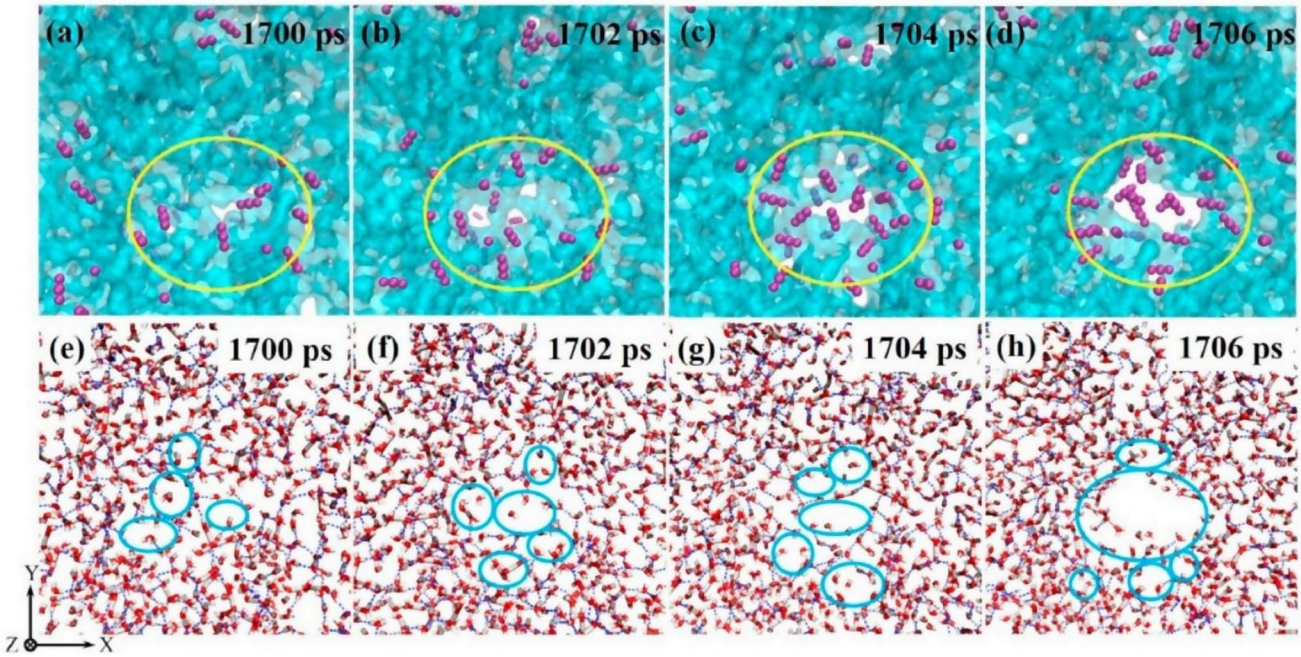


Fig. 9 Details (top View) of carbon dioxide diffusing into the water film at (**a**) t ¼ 1700 ps, (**b**) t ¼ 1702 ps, (**c**) t ¼ 1704ps, and (**d**) t ¼ 1706 ps, and snapshots (top View) of hydrogen bonds in the water film at (**e**) t ¼ 1700 ps, (**f**) t ¼ 1702 ps, (**g**) t ¼ 1704 ps and (**h**) t ¼ 1706 ps. The blue circles highlight where the hydrogen bonds have been broken and the hydrogen bond is highlighted as a purple dashed line between the oxygen (in red) and hydrogen (in white). Color codes for atoms are, purple: CO_2; cyan: water; grey: octane; yellow, silicon; red, oxygen; and white, hydroxyl hydrogen (Hh) [69]

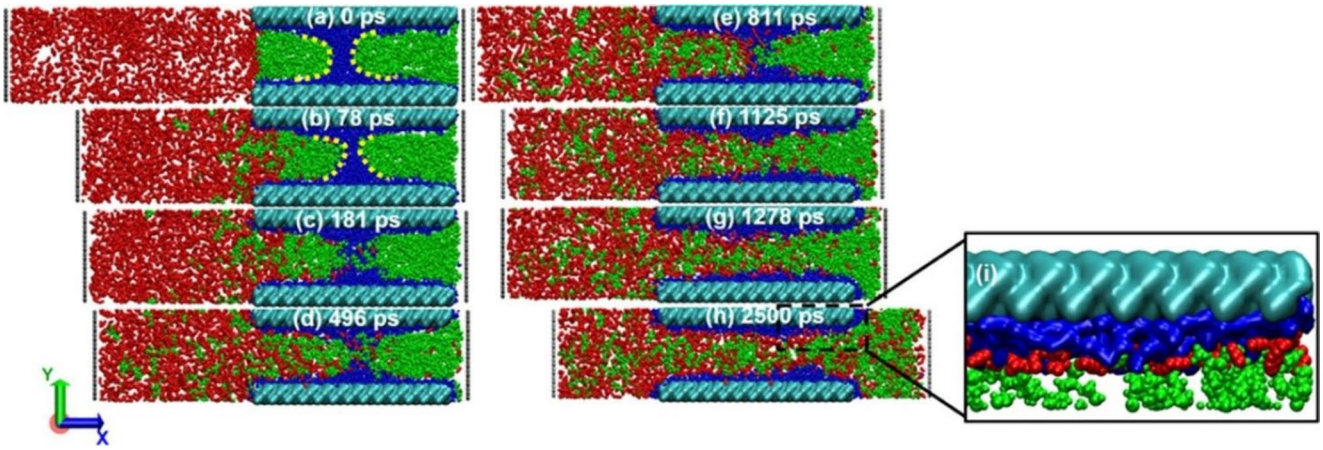


Fig. 10 Snapshot of CO_2 breaking through the water bridge at typical moments: (**a**) 0 ps, (**b**) 78 ps, (**c**) 181 ps, (**d**) 496 ps, (**e**) 811 ps, (**f**)1125 ps, (**g**) 1278 ps, and (**h**) 2500 ps. The configuration of water/octane/CO_2 molecules near the upper surface at 2500 ps is enlarged in figure (**i**). The yellow dashed lines in (**a**) and (**b**) were plotted to more clearly describe the shape of water bridge−octane interfaces. CO_2/ water/oil/wall surfaces/graphene sheets are represented by red/blue/green/cyan/black, respectively [70]

the system evolves spontaneously toward a low-free-energy state, achieving efficient oil displacement and CO_2 sequestration [70] (Fig. 10).

Li (2022) used alternating-injection experiments on low-permeability sandstone and related analyses to examine the microscale migration and interaction of CO_2 and formation water. CO_2 dissolves in formation water to form carbonic acid, which causes dissolution-induced permeability enhancement and ion migration at the pore scale, while CO_2 also enters small pore throats by molecular diffusion; active water regulates interfacial tension and acts with the aqueous phase and CO_2 to mobilize pore fluids uniformly and delay gas breakthrough. This clarifies the migration patterns of CO_2 and formation water in pores and their synergistic oil-displacement effect [71]. Wang et al. (2024) used the lattice Boltzmann method to simulate CO_2–oil–water three-phase displacement and examine the microscale mechanisms of CO_2-based oil displacement and sequestration.

CO_2 preferentially displaces the water phase along large pores, forming dominant pathways; higher initial water saturation gives higher sweep efficiency and sequestration effectiveness. Mixed-phase displacement improves both oil recovery and sequestration, whereas non-mixed-phase displacement is less efficient. Narrow pore throats in the rock matrix are a key factor limiting residual-oil mobilization and CO_2 sequestration; optimizing injection and production can improve both outcomes [72]. Han et al. (2025) performed supercritical CO_2 displacement experiments on a long-core sample from the Mo109 reservoir under simulated geothermal and pressure conditions, focusing on the migration and interaction of CO_2 and formation water. CO_2 dissolves in formation water to form carbonic acid, triggering a dissolution reaction that widens pore throats; CO_2 also penetrates small pores by molecular diffusion and interacts with crude oil. Among the displacement methods tested, WAG shows the best flow-control characteristics, improving fluid migration efficiency and oil recovery [73]. Xia et al. (2025) built a CO_2–water film–crude oil–rock molecular system and used molecular dynamics simulations to observe fluid-molecule distribution, mineral-interface adsorption, and crude-oil diffusion under different water-film thicknesses, examining how the water film regulates CO_2 mass transfer and crude-oil properties at the molecular scale. A continuous water film on the rock wall occupies the pore-edge space and compresses crude-oil molecules toward the narrow center of the pore channel; combined with the oil–water interfacial force and the restraining effect of the hydrogen-bonded water network, hydrocarbon molecules become more densely arranged, the number of molecules per unit volume rises, the equivalent density of crude oil increases by 86.9%, and the diffusion coefficient falls to 72.30% of its original value. The inherent confinement of the nanopores further intensifies the aggregation and compression of crude oil, lowering its flowability. The complete water film acts as a physical barrier and is the main obstacle to CO_2 penetration into crude oil; thicker water films inhibit CO_2 diffusion more strongly, and CO_2 adsorption density on the rock surface decreases markedly with film thickness. Compared with pressure, heating more effectively disrupts the hydrogen-bonded structure of the water film, enhances CO_2 mass transfer and diffusion, and improves crude-oil flowability [74] (Fig. 11). Gandomkar et al. (2025) used interfacial-tension (IFT) measurements to show that a CO_2-soluble nonionic surfactant (2EH-PO_5-EO_{15}) raises CO_2 solubility by about 100% (from 1.25 to 2.51 mol%) through micelle-mediated solubilization. Through interface-directed arrangement, the IFT was lowered from 32.4 to 6.7 dyn/cm, allowing stable foam to form (> 24 h); through surfactant adsorption, the rock wettability shifted from strongly water-wet (18°) to intermediate gas-wet (71°) [75].

Although current research on the microscale mechanisms of CO_2–water interaction has advanced considerably, several gaps remain. The functional role of the water film is disputed: one view treats its rupture as a prerequisite for CO_2 to enter blind pores and displace oil efficiently, whereas the other regards an intact water film as the main barrier to CO_2 permeation. No unified quantitative criterion has yet been established for the critical conditions under which the water film shifts from a barrier to a channel. The dominant controls on water-film rupture are also unresolved, and the relative importance and applicable conditions of three mechanisms—CO_2 dissolution disrupting the hydrogen-bond network, high-speed scouring and dragging, and temperature-induced weakening of hydrogen bonds—remain unclear. Existing simulations rely mainly on idealized systems and overlook the effect of high-mineralization ions on water-film stability. Future work should establish unified criteria, develop multi-factor coupled and cross-scale methods, and broaden active intervention by surfactants, moving the water film from passive breakthrough to active regulation.

3.2 Microscopic Inhibition of CO_2 displacement by the aqueous phase

CO_2 flooding is prone to gas channeling, which leads to unstable displacement; the water phase in the reservoir can suppress gas channeling at the microscale by dissolving and consuming the gas phase, occupying pore channels, and changing interfacial properties, which stabilizes the displacement front. Most existing studies, however, focus on single-phase dissolution, and the microscale mechanisms by which the water phase suppresses CO_2 channeling remain unclear. The intrinsic mechanisms of water-phase stabilization of displacement therefore need to be examined on the basis of CO_2–oil–water phase equilibrium and dissolution behavior, providing theoretical support for optimizing CO_2 flooding [76].

Zhang et al. (2019) used molecular dynamics simulations to examine crude-oil diffusion in different rock pores under water-bearing conditions, considering both molecular diffusion and interfacial adsorption. The hydrophobicity of organic surfaces, together with strong hydrogen bonding among water molecules, promotes water aggregation at the centers of graphite-like pores and strongly inhibits crude-oil diffusion; this reveals the microscale mechanism by which the spatial distribution of the aqueous phase impedes oil–gas mass transfer [77]. Chowdhury et al. (2023) used microfluidic experiments to examine pore-scale interactions among three-phase fluids during CO_2-WAG injection. The interfacial behavior and migration pathways of water, oil, and CO_2 were observed in real time in a micromodel, clarifying

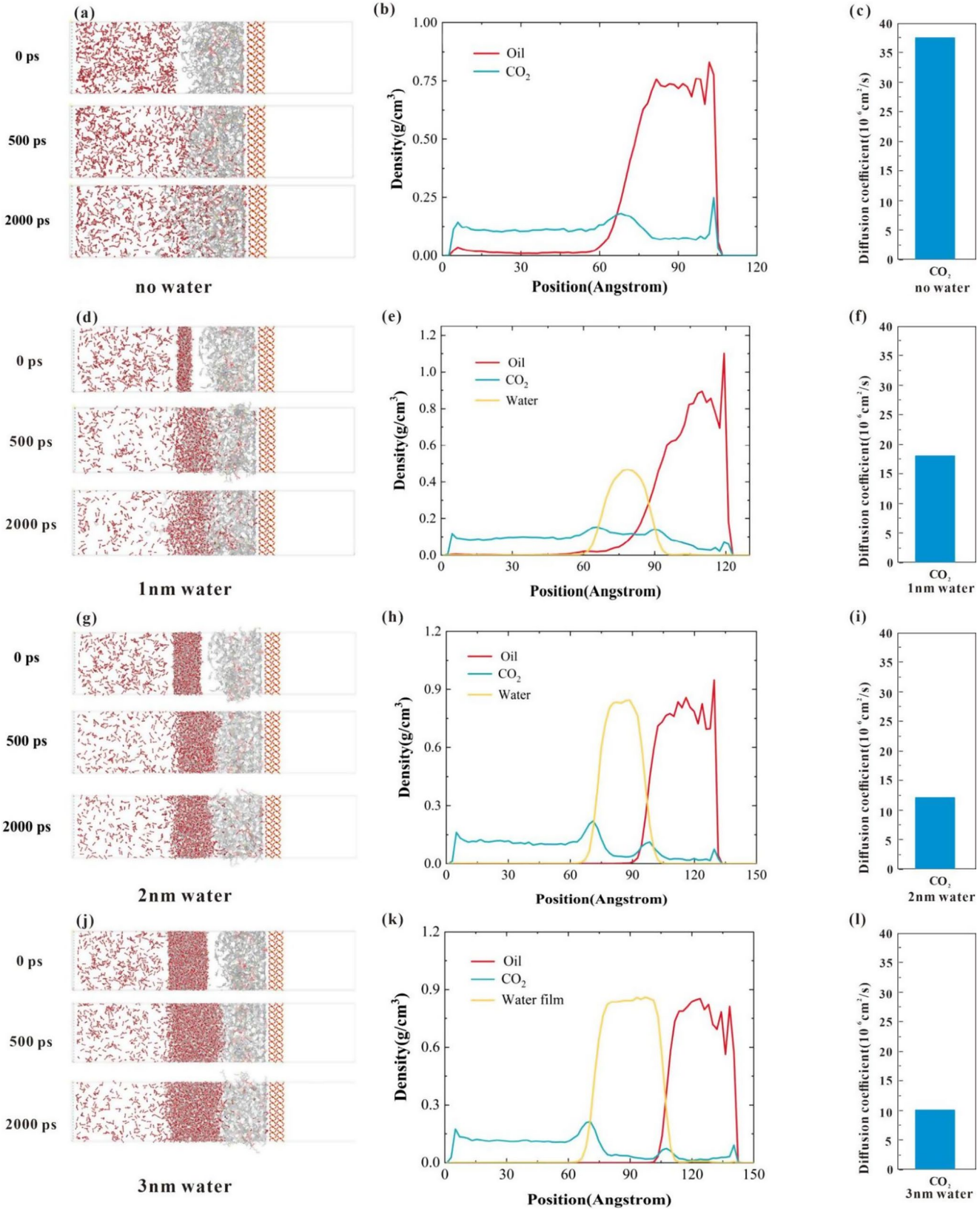

(a)
0 ps
500 ps
2000 ps
no water
(b)
Density(g/cm3)
Position(Angstrom)
Oil
CO2
(c)
Diffusion coefficient(10-6cm2/s)
CO2
no water
(d)
0 ps
500 ps
2000 ps
1nm water
(e)
Density(g/cm3)
Position(Angstrom)
Oil
CO2
Water
(f)
Diffusion coefficient(10-6cm2/s)
CO2
1nm water
(g)
0 ps
500 ps
2000 ps
2nm water
(h)
Density(g/cm3)
Position(Angstrom)
Oil
CO2
Water
(i)
Diffusion coefficient(10-6cm2/s)
CO2
2nm water
(j)
0 ps
500 ps
2000 ps
3nm water
(k)
Density(g/cm3)
Position(Angstrom)
Oil
CO2
Water film
(l)
Diffusion coefficient(10-6cm2/s)
CO2
3nm water

Fig. 11 (**a**) Molecular dynamics simulation of system configuration snapshots without water film; (**b**) Density profiles of individual components without water film; (**c**) Diffusion characteristics of crude oil without water-film thicknesses; (**d**) Molecular dynamics simulation of system configuration snapshots at 1 nm water film; (**e**) Density profiles of individual components at 1 nm water film; (**f**) Diffusion characteristics of crude oil at 1 nm water film; (**g**) Molecular dynamics simulation of system configuration snapshots at 2 nm water film; (**h**) Density profiles of individual components at 2 nm water film; (**i**) Diffusion characteristics of crude oil at 2 nm water film; (**j**) Molecular dynamics simulation of system configuration snapshots at 3 nm water film; (**k**) Density profiles of individual components at 3 nm water film; (**l**) Diffusion characteristics of crude oil without at 3 nm water film [74]

how the aqueous phase splits flow and controls gas movement. The aqueous phase generates the Jamin effect at pore throats, suppressing CO_2 channeling while regulating CO_2 diffusion and dissolution pathways in the pores, which strengthens CO_2–crude-oil miscible displacement [78]. Zhao et al. (2023) used homogeneous/heterogeneous core flooding with CO_2-WAG and numerical simulations and found that the aqueous phase preferentially enters high-permeability throats, forming plugging and flow regulation in the pores that reduces CO_2 channeling velocity; the aqueous film also changes the gas–oil interface distribution and enlarges the contact area between CO_2 and crude oil. Stronger reservoir heterogeneity makes water injection more effective at controlling channeling [79]. Lu et al. (2024) used molecular dynamics simulations to examine CO_2-driven oil recovery in water-bearing blind-ended nanopores. Based on the mechanism by which CO_2 dissolution disrupts the hydrogen-bond network of the water film, crude-oil displacement begins only after CO_2 dissolves in water, disrupts the hydrogen-bond network of water molecules, and ruptures the water film; this clarifies the microscale control that water-film stability exerts on the initiation of CO_2-based oil displacement [80]. Zhao et al. (2024) performed alternating CO_2–modified water displacement experiments under simulated formation temperature and pressure using sand-filled tubing and Berea cores, combined with NMR T_2 spectroscopy to examine the microscale displacement mechanism. CO_2 mobilizes microporous crude oil, while modified water activates macroporous crude oil; alternating injection of the two fluids suppresses gas channeling, achieves efficient displacement across all pore sizes, identifies optimal injection parameters, and gives an alternating-displacement recovery 15.32% higher than water flooding alone [81].

Qin et al. (2025) performed a three-dimensional multicomponent CO_2–water–oil multiphase simulation at the shale fracture-pore scale. Using the volume-of-fluid (VOF) method coupled with miscible convection–diffusion and immiscible-flow models, they controlled injection velocity and wettability and characterized interfacial tension, capillary force, and contact-angle effects. Water-alternating- CO_2 displacement uses capillary force to stabilize the displacement front, enhances CO_2 dissolution and sweep efficiency, and exploits strong hydrophilicity to promote both oil recovery and carbon sequestration, revealing the microscale mechanisms by which the aqueous phase inhibits channeling and strengthens mass transfer [82]. Xu et al. (2025) performed water–gas alternating displacement experiments on conglomerate cores of varying permeability. Based on micro-flow and interfacial-interaction principles that reduce CO_2 mobility in the aqueous phase, the effect of permeability on water–gas alternating-flow control was analyzed: as permeability increases, the ability of alternating flow to build resistance weakens, preventing the aqueous phase from effectively sealing CO_2 channeling pathways; the aqueous phase also continuously diminishes the microscale extraction reaction between CO_2 and crude oil, reducing the enhanced-oil-recovery effect of water–gas alternating processes in high-permeability cores [83]. Zhang et al. (2026) performed microscale control experiments on CO_2–polymer solution plugback flooding. Using a heterogeneous three-tube core model, they simulated microscale plugging and phase transition of polymers and CO_2 in water-flooded high-permeability zones. Polymer molecules form a network in the pore water of the high-permeability layer, plugging large water-conducting channels and forcing CO_2 and the aqueous phase to migrate toward low-permeability pores. After CO_2 dissolves in water, it slightly lowers the intrinsic flow resistance of the aqueous phase at the molecular level, but the plugging effect of the polymer in the high-permeability channel dominates, driving subsequent injected fluid into the more resistive low-permeability pore network. Macroscopic injection pressure rose by 3.2 MPa, recovery from the low-permeability zone increased by 37.10%, and the final recovery factor reached 57.4% [84].

Current research on aqueous-phase-controlled CO_2 flooding has identified many microscale mechanisms, but key controversies remain at both the mechanistic-decoupling and engineering-applicability levels. The dual nature of the aqueous phase is still unresolved: the Jamin effect and capillary force can control CO_2 migration, but the water film blocks oil–gas mass transfer and severely limits diffusion and dissolution, so no unified quantitative criterion exists for the critical conditions under which the aqueous phase shifts from migration control to mass-transfer inhibition. The effectiveness of aqueous-phase migration control is also strongly reservoir-dependent: as permeability rises, the ability of alternating water–gas flow to build resistance weakens, the aqueous phase can no longer seal CO_2 migration pathways, and CO_2 extraction of crude oil declines, leaving the adaptive boundaries of WAG unclear. Future work should establish a unified criterion for the transition between the dual properties of the water film and a multifactor quantitative analysis framework, develop multi-field

coupled numerical simulation and cross-scale prediction methods, and move aqueous-phase-controlled CO_2 flooding from qualitative understanding to quantitative design.

4 Research status of CO_2–rock microscopic reaction mechanisms

4.1 CO_2 interfacial adsorption and mineral acidification–dissolution effects

4.1.1 Microscopic confinement adsorption behavior of CO_2 in nanopores

In geological CO_2 sequestration, spatial confinement alters the microscopic mechanisms of CO_2 adsorption and strongly influences both sequestration capacity and safety [93]. Confined CO_2 adsorption in nanopores directly determines physical sequestration efficiency and is closely coupled with mechanisms such as residual trapping and solubility trapping [85, 86]. Elucidating the microscopic behavior of confined adsorption provides a basis for assessing the sequestration potential of low-permeability reservoirs and optimizing injection–production parameters, which contributes to the long-term stability of geological CO_2 sequestration [85].

Wang et al. (2016) used molecular dynamics simulations to examine crude-oil migration through inorganic nanopores in shale. Using a quartz slit as the model system and the COMPASS force field to calculate solid–liquid molecular interactions, they analyzed adsorption, diffusion, and interfacial potential energy for CO_2, oil, and water. CO_2 preferentially adsorbs onto the quartz surface and displaces crude oil; through van der Waals interactions, CO_2 disrupts oil–rock adsorption, lowers viscosity and interfacial tension, alters wettability, and improves oil displacement efficiency [86]. Yang et al. (2017) used molecular dynamics simulations to examine the adsorption of water and octane on pore surfaces of different rocks. Comparing the adsorption energies of water and octane on quartz and graphite pores, they found that octane adsorption on both graphite and quartz was much stronger than water adsorption, and that the graphite–octane interaction was stronger still. Rock-surface wettability determines the adsorption distribution of both the water film and crude oil, providing an interfacial-adsorption basis for understanding how the water film affects the CO_2–rock–crude-oil interface [87]. Zhang et al. (2019) used molecular dynamics simulations to examine crude-oil diffusion in shale nanopores under water-bearing conditions. Building a molecular model of water-containing shale nanopores, they analyzed how rock-surface wettability and hydrogen bonding among water molecules affect crude-oil diffusion. The hydrophobic nature of organic rock surfaces and strong hydrogen bonding among water molecules cause water to aggregate at the center of graphite-like pores, forming a water-film barrier that strongly inhibits crude-oil diffusion; this clarifies, from a microscale interfacial perspective, how the water film obstructs crude-oil migration [77]. Liu et al. (2020) performed experiments on rock wettability transition under low-salinity water flooding. Using interface characterization and core displacement, they examined how a shift in rock-surface wettability from oil-wet/hydrophilic to neutral affects the initiation and migration of residual oil. Neutralization of rock-surface wettability promotes desorption of crude oil from the rock surface and its subsequent migration in pore throats; wettability alteration is thus the core microscale mechanism by which water-based systems enhance oil recovery through interfacial interactions [88]. Zhang et al. (2022) performed high-temperature, high-pressure microfluidic experiments using a microfluidic chip with rock-pore structures to simulate deep-reservoir conditions and observe CO_2–oil miscible displacement in real time. By reproducing the microscale channel structures of rock pores and fractures, they found that CO_2 extracts light crude-oil components in a funnel-like pattern, penetrates water films in rock pores to reach blind zones containing residual oil, and is accompanied by asphaltene deposition; this clarifies the microscale mechanisms of CO_2 extraction and interfacial breakthrough in rock pore channels [89].

Gandomkar et al. (2023), based on three-phase interface experiments on carbonate rocks, state that prior to wettability characterization, cores must undergo constant-temperature static aging at formation temperature and pressure for 1–2 days to reach equilibrium. Sufficient aging allows competitive adsorption of asphaltenes, resins, and formation-water ions onto mineral surfaces, establishing an oil–water interface that reflects the true reservoir characteristics. Inadequate aging can cause uneven fluid distribution within the pores, prevent asphaltene adsorption from reaching a steady state, reduce the accuracy of contact-angle measurements, and make it difficult to evaluate the regulatory effects of CO_2, toluene, and nanoscale additives on wettability reversal and asphaltene deposition. In laboratory experiments, graded aging protocols—24 h, 7 d, and 14–30 d—are used under simulated operating conditions, with all core samples subjected to identical aging conditions; this approach eliminates experimental errors arising from differences in interfacial adsorption and ensures stable, reliable wettability-characterization results [90]. Lü et al. (2023) examined oil displacement efficiency in quartz, illite, and smectite cores using nuclear magnetic resonance (NMR) combined with CO_2 displacement experiments. Quantifying oil recovery in cores with different mineral compositions and pore sizes, they found that illite-bearing cores gave

the highest recovery and best displacement performance; smectite-bearing cores displaced oil well in large pores but poorly in small pores, giving the lowest overall recovery. These results confirm that mineral type regulates pore-scale displacement [91]. Wang et al. (2023) performed pore-scale simulations of shale oil–CO_2–water three-phase flow, using a multi-component pseudopotential model to simulate miscible displacement in the presence of a water phase. The Jamin effect and capillary resistance were included to characterize the barrier effect of the water phase in rock pores; the water phase occupies the center of the rock pore, and capillary resistance hinders miscible mass transfer between CO_2 and oil, enriching residual oil and reducing CO_2 sequestration capacity. These findings clarify the microscale mechanism by which coupling between rock-pore structure and water-phase distribution inhibits miscible displacement [92]. Chen et al. (2026) used X-ray diffraction, scanning electron microscopy, and energy-dispersive spectroscopy to show that feldspars (albite and K-feldspar) dissolve preferentially; enlarged pores and throats increase the specific surface area available for contact between CO_2 and reservoir fluid, raising CO_2 solubility and mineral-trapping sequestration potential. As interaction time increases, dissolution of rock-surface minerals, formation of kaolinite, and increased surface roughness make the rock more hydrophilic; this enhanced hydrophilicity improves residual-gas (capillary) sequestration, allowing more CO_2 to be trapped stably in the pore space as isolated bubbles or thin films and reducing leakage risk [93] (Fig. 12).

Existing research on CO_2–rock interface interactions still faces several controversies at both the mechanistic and engineering-applicability levels. The role of wettability is disputed: one view treats hydrophilic conditions as favorable for oil recovery, while the other emphasizes water-film interference; the former helps CO_2 displace oil films, whereas the latter inhibits oil–gas mass transfer, and no unified criterion exists for the pore-throat size or water saturation that governs this transition. Conclusions on the regulatory effect of mineral type also vary: the marked difference in displacement efficiency between illite and smectite underscores the influence of mineral composition on pore-scale utilization, but most current findings come from idealized single-mineral models and do not account for competitive adsorption and differential dissolution in multi-mineral systems, limiting their applicability to real reservoirs. Future work should establish a unified transition criterion based on the dual nature of wettability, examine competitive adsorption and cross-scale coupling in multi-mineral systems more deeply, and move the understanding of CO_2–rock interface interactions from qualitative insight toward quantitative design.

4.1.2 Microscopic mechanisms of multi-mineral acidification and dissolution

The interaction between acidic fluids and multi-mineral systems (igneous and sedimentary rocks) is a core scientific issue in geological carbon sequestration and oil-and-gas reservoir alteration, and its dissolution behavior directly determines mineral transformation pathways and reservoir porosity–permeability evolution. Fracture aperture and surface roughness strongly affect the dissolution–precipitation balance by regulating interfacial mass transfer and local chemical fields; small apertures and high surface roughness promote iron-oxide precipitation, intensify surface passivation, and lower dissolution efficiency. Most existing studies focus on single-mineral dynamics and overlook the coupled effects of micromorphology, interfacial mass transfer, and passivation, leaving the microscale mechanisms of multi-mineral acid-induced dissolution incompletely understood. Quantitative mechanistic studies are needed to support engineering applications [94].

Tang et al. (2016) performed experiments on mineral dissolution and migration during CO_2 injection into the water-flooded formation of the Pucheng Oilfield, simulating the microchemical reactions between injected CO_2 and rock minerals and observing the migration of mineral elements and their morphological changes. CO_2 injection induces selective dissolution of rock minerals, with some minerals undergoing precipitation transformation; at the microscale this alters the pore structure and surface composition of the rock, affecting oil recovery in the reservoir [95]. Matter et al. (2016) conducted field-scale and laboratory simulation experiments on basalt CO_2 mineralization at the CarbFix project in Iceland. Based on field injection monitoring and ^{14}C tracer studies, combined with mineralogical analysis to characterize mineralized products and reaction rates, they confirmed that basalt can achieve over 95% CO_2 mineralization and sequestration within two years. The reaction rate first peaks and then gradually decreases and stabilizes, consistent with the double-exponential kinetics observed in the laboratory; this reveals the microscopic mechanism of permanent sequestration, involving rapid dissolution of magnesium-rich ferrosilicates and stable precipitation of carbonates [96]. Lin et al. (2022) performed multi-phase interaction experiments between CO_2, water, and tight sandstone, controlling injection pressure and temperature, observing the evolution of porosity and permeability, and integrating microscopic mineral dissolution and precipitation to analyze the microscale mechanisms behind changes in rock properties. When injection pressure increased from 15 MPa to 25 MPa, the porosity and permeability of the tight sandstone rose by 3.2% and 9.9%, respectively; when temperature rose from 44 °C to 64 °C, the porosity increase

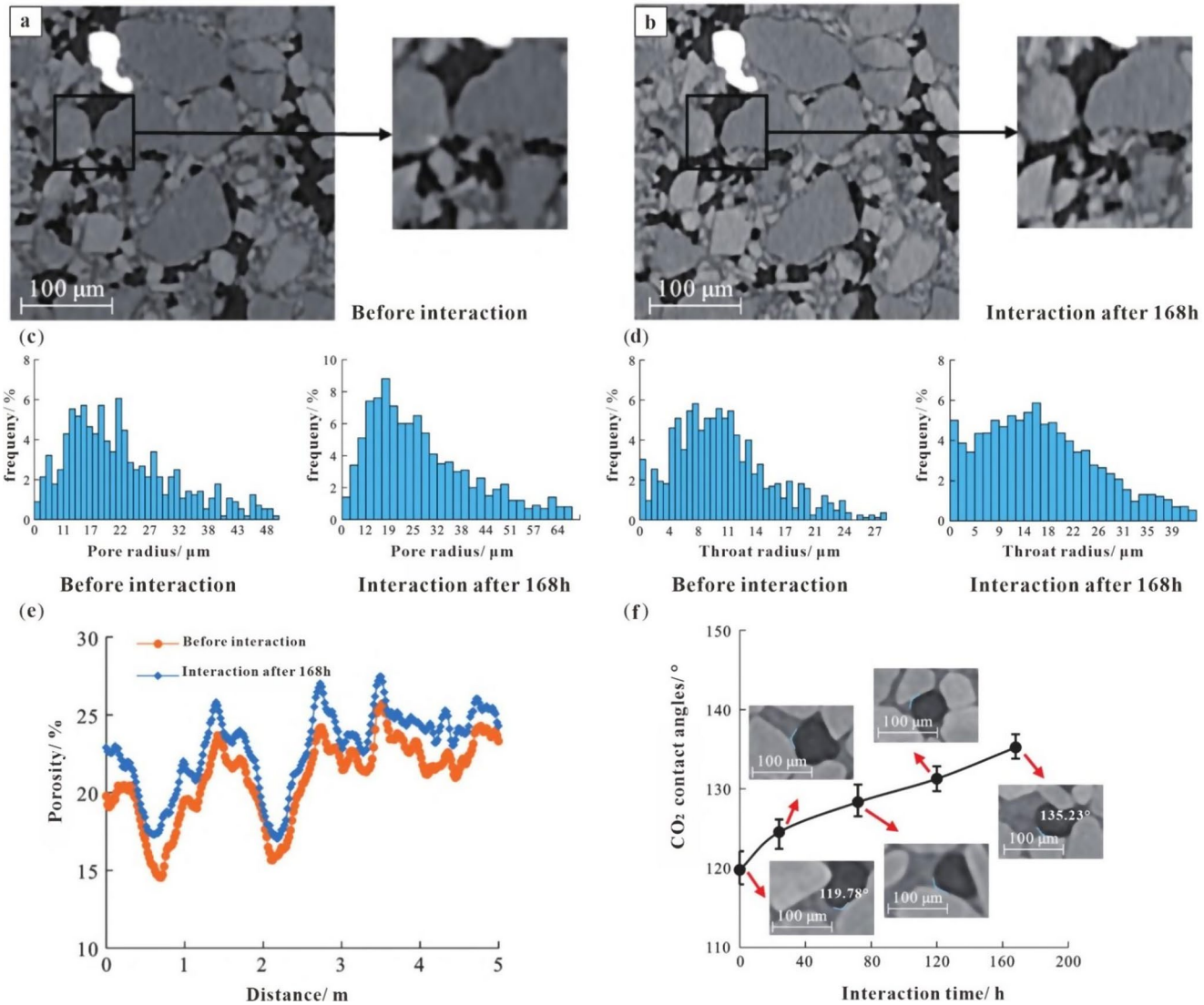


Fig. 12 Microscopic evolution of sandstone reservoir properties induced by CO_2-brine-rock interactions during CO_2 flooding and storage. (**a**) Initial microscopic morphology of sandstone pore structure before interaction; (**b**) Pore structure morphology after 168 h of CO_2-brine-rock interaction; (**c**) Changes in pore radius distribution before and after interaction; (**d**) Changes in throat radius distribution before and after interaction; (**e**) Longitudinal porosity distribution along the core before and after interaction; (**f**) Evolution of CO_2-water contact angle with interaction time [93]

fell from 3.0% to 1.7%. At the microscale, CO_2-formed carbonic acid first dissolves the carbonate cement in the sandstone, followed by mineral recrystallization, which regulates pore-throat development [97]. Liu et al. (2022) performed supercritical CO_2 distillation–water soaking experiments on shale samples from the Qing-1 Formation in the Songliao Basin to simulate CO_2–water–rock interaction under geological conditions, focusing on the microscale mechanisms of multi-mineral acidification and dissolution. Supercritical CO_2 combined with water forms an acidic fluid that dissolves carbonate minerals (calcite and dolomite) in the shale, generating large pores; this process also promotes contraction of clay minerals, forming microfractures that enhance rock porosity and permeability and enable long-term CO_2 sequestration [98]. Tao et al. (2023) performed CO_2–shale interaction experiments under different water-saturation conditions to examine the microscale mechanisms of CO_2-based oil recovery and sequestration. Under high water saturation, CO_2 dissolves minerals and increases pore volume and throat diameter, improving both oil displacement efficiency and sequestration capacity; under low water saturation, particles clog pore throats, reducing both oil recovery and sequestration efficiency; under no-water conditions, only organic matter is extracted, and rock properties change little. Water saturation is therefore the key factor controlling the effectiveness of CO_2-based

oil recovery and sequestration [99]. Dong et al. (2024) performed dynamic dissolution experiments using limestone with CO_2-saturated brine to simulate formation water–rock reactions during CO_2-enhanced oil recovery (CO_2-EOR) and carbon sequestration. After 60 days of immersion in CO_2, the uniaxial compressive strength and elastic modulus of the limestone decreased by 79.3% and 72.03%, respectively. At the microscale, CO_2 dissolves carbonate minerals and disrupts the cementing structure, weakening the rock skeleton and lowering the fracture resistance of the reservoir or caprock, which threatens the stability of the CO_2-EOR reservoir and the long-term safety of CO_2 sequestration [100]. Zhang et al. (2024) performed adsorption and reaction experiments on CO_2–water–rock systems for shale-oil reservoirs, targeting CO_2-EOR and sequestration scenarios. CO_2 undergoes interfacial adsorption and chemical reactions with minerals and fluids in the shale nanopores, collectively altering pore-throat structure and flow conditions; CO_2 preferentially reacts at the microscale with feldspars and clay minerals in the shale, causing dissolution that enlarges the nanopores, accompanied by precipitation of small amounts of silica and clay. The process is dominated by pore enlargement, providing a microscale mechanistic basis for CO_2-based enhanced oil recovery and geological sequestration in shale-oil reservoirs [101]. Gao et al. (2025) performed CO_2–formation water–dense sandstone reaction experiments to simulate formation conditions at 89 °C and 37.0 MPa. CO_2 dissolved in water forms a weakly acidic fluid that induces dissolution of minerals such as feldspars; over time, dissolution-induced pores and cavities formed on the rock surface [102]. Zhang et al. (2025) performed long-term CO_2–water–rock reactions under high-temperature, high-pressure conditions, simulating the core displacement process under geological conditions (100 °C, 24 MPa); mineral evolution was characterized by ion chromatography, SEM, and XRD. CO_2 dissolves in water to form a weakly acidic fluid that induces mineral dissolution and recrystallization. Calcite dissolution was most pronounced in the sandstone, while K-feldspar dissolution increased the K^+ concentration in solution; secondary kaolinite and clay particles also migrated and blocked pore throats, reducing core permeability [103].

In core displacement experiments, the capillary end effect at the outlet can distort the saturation distribution, interfering with accurate evaluation of recovery factor and relative permeability. Rapoport and Leas proposed the classical scaling criterion (L, V, μ) based on the competition between viscous and capillary forces; when the product of core length (L), displacement velocity (V), and displacement-phase viscosity (μ) exceeds a critical threshold, viscous forces dominate and the end effect can be neglected. This criterion has been widely used in experimental design—for example, Gandomkar et al. set the scaling coefficient to 7.5 in CO_2 foam flooding to stabilize the displacement front. Nematzadeh et al. noted, however, that even after satisfying this criterion, the pressure-drop signal may still be obscured by the relative-permeability lag effect, making the breakthrough point difficult to identify. In studies of the microscale mechanisms governing CO_2–oil–water–rock systems, the Rapoport–Leas criterion should therefore be the basis for optimizing injection rates and core dimensions, while the superimposed interference of the end effect and the lag effect on the pressure-drop signal must be distinguished to improve the reliability of quantitative correlations between microscale parameters and macroscopic engineering indicators [104–106]. Maniar et al. (2025) performed long-term CO_2-saturated brine experiments under in situ stress on mudstone samples from the Vagn Formation in the Central Graben of Denmark, using both low-frequency forced oscillation and high-frequency pulse transmission to monitor elastic and acoustic parameters synchronously. After CO_2 exposure, rock parameters decreased by 3%–6%, while Young's modulus dispersion ranged only from 3% to 11%. At the microscale, these changes were attributed to minor mineral dissolution and microfracture expansion induced by acidizing fluids; the mudstone remained structurally stable and showed long-term CO_2-sequestration sealing capacity [107]. Yang et al. (2026) performed SEM morphological observations of feldspar under CO_2-sequestration conditions. After CO_2 injection, feldspar dissolved predominantly, with sodalite dissolving more strongly than K-feldspar and forming intragranular dissolution pores accompanied by secondary precipitation of kaolinite; carbonate minerals (calcite, dolomite, siderite) dissolved markedly, showing rough crystal surfaces and enlarged pores, releasing cations and promoting CO_2 mineral trapping, which improved reservoir porosity and permeability. The total clay-mineral content decreased, with smectite and chlorite dissolving preferentially; kaolinite first decreased and then increased as a result of feldspar alteration; secondary clay minerals tended to clog micropore throats, causing a marked decline in permeability in high-clay-content reservoirs [108]. Wei et al. (2020) used scanning electron microscopy (SEM) to compare the micromorphology of dense sandstones before and after exposure to supercritical CO_2, observing mineral–fluid interface reactions. Feldspar minerals are etched by acidic fluids, showing rough crystal surfaces with dissolution grooves developed along cleavage planes. Carbonate minerals including calcite, dolomite and siderite suffer intensive dissolution, featuring rounded crystal edges and abundant dissolution pits and micropores that effectively enhance inter-particle porosity; dolomite exhibits weaker dissolution degree than calcite, and siderite can be subjected to clay alteration to form kaolinite. For

clay minerals, kaolinite displays increased aggregation, and pore-throat blockage tends to occur in high-content zones. The original honeycomb-like and sheet-shaped structures of illite/smectite and chlorite/smectite mixed-layers are damaged. Micropores increase while pore connectivity deteriorates. Overall, co-existing effects of dissolution-driven pore enlargement and pore-throat clogging caused by clay migration can be observed [5] (Fig. 13).

Despite existing research on CO_2-induced acidification and dissolution of multi-mineral rocks (Table 1), several controversies remain. First, dissolution response varies markedly among lithological reservoirs: basalt exhibits very high mineralization efficiency, whereas limestone undergoes considerable mechanical-strength reduction; yet the factors controlling these differences remain unclear. Second, studies disagree on the dual "permeability enhancement" and "confinement" effects of mineral dissolution, in which pore expansion from dissolution and secondary precipitation causing throat blockage occur simultaneously; the critical temperature–pressure conditions and mineral assemblages at which this dissolution–precipitation competition shifts from permeability-enhancement dominance to confinement dominance remain unquantified. Future work should identify the factors controlling differential dissolution among lithologies, quantify the critical temperature–pressure conditions and mineral assemblages that trigger the shift from "permeability enhancement" to "confinement" in the dissolution–precipitation competition, develop long-term evolution prediction models that incorporate multi-mineral coupling, and move CO_2–rock acidification and dissolution studies from descriptive observation toward quantitative prediction.

4.2 Microscopic mechanisms of secondary mineral evolution under ion migration

The ion migration rate, interfacial mass transfer, and their coupled regulation of mineral nucleation, phase transformation, and morphological evolution directly determine sequestration stability and capacity. The microscale interfacial mechanisms behind ion-migration-driven mineral evolution and the associated kinetic competition remain unclear; in particular, the synergistic interaction between hydrate formation and carbonate mineralization has not been characterized quantitatively [109]. Clarifying the intrinsic links between ion migration and secondary-mineral evolution offers useful theoretical support for long-term safety assessment and engineering optimization of carbon sequestration systems.

Wang et al. (2013) performed CO_2–water–rock interaction simulation experiments combined with SEM–EDS characterization to analyze the microstructure and elemental composition of new phases, identifying mineral-trapping intermediate products. At 55 °C, no new minerals formed, whereas at 70 °C, 85 °C, and 100 °C, needle-shaped, spherical, and flower-like new minerals appeared on the sample surfaces, respectively; these are key intermediate products in CO_2 mineral sequestration [110] (Fig. 14). McGrail et al. (2017) performed both field and laboratory experiments on the basalt–supercritical CO_2–brine reaction. Based on the acidification principle whereby CO_2 dissolves in water to form carbonic acid, they conducted in situ injection and batch-reaction tests on a basalt reservoir in the Columbia Plateau. At the microscale, native minerals in the basalt—such as glassy material, monobahirite, and plagioclase—were dissolved by carbonic acid, releasing ions such as Ca^{2+}, Mg^{2+}, and Fe^{2+}, which reacted rapidly with dissolved CO_2 to form carbonate minerals like dolomite and calcite, achieving CO_2 mineralization sequestration [111]. Li et al. (2020) performed multi-field coupled numerical simulations of CO_2–water–rock systems in saline aquifers. Using a thermal–fluid–mechanical coupling model, they simulated pore-fluid flow, heat transfer, and rock-stress evolution during CO_2 injection and production, examining the microscale mechanisms of water–rock chemical reactions triggered by CO_2 dissolution: CO_2 dissolves in formation water to form carbonic acid, which ionizes H^+ and dissolves minerals such as feldspar and calcite, releasing ions like Ca^{2+} and Mg^{2+}; clay minerals also swell on contact with water, altering pore-throat structure and fluid-flow pathways at the microscale [112]. Ma et al. (2020) performed one-dimensional reactive-transport simulations and lithofacies experiments on deep sandstones in the Bohai Bay Basin, using mineral-kinetics principles to examine the microscale interaction mechanisms of CO_2, water, and rock. CO_2-acidified fluids promote dissolution of calcite and chlorite, releasing ions while facilitating their conversion to dolomite and sideromelane; the slow dissolution of aluminosilicates is the rate-limiting step for carbonate precipitation, improving overall pore-throat connectivity without significant blockage and confirming that this system can achieve stable CO_2 mineral sequestration [113].

Sun et al. (2021) performed comparative experiments on the interfacial rheology of nanoparticle foams and their retention in porous media, based on particle adsorption, pore-throat blockage, and interfacial-film strengthening, to quantify nanoparticle retention in the rock matrix. Nanoparticle retention decreases as rock permeability increases; through interfacial-rheological enhancement of foam stability, the nanoparticles form bridging and sealing structures at the pore-throat scale, selectively reducing CO_2 mobility in high-permeability pathways, improving sweep efficiency in heterogeneous reservoirs, and facilitating long-term CO_2 sequestration [114]. Song et al. (2023) performed microscale

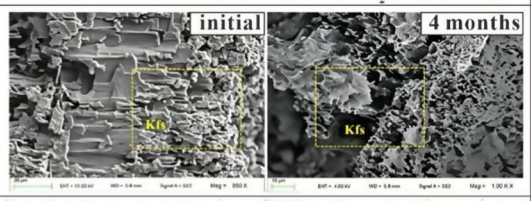
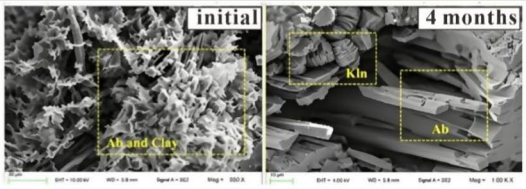
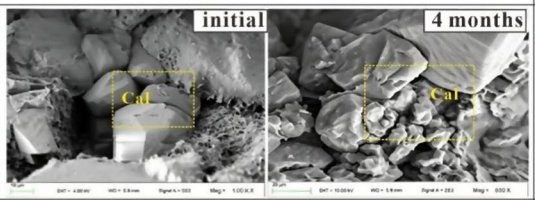
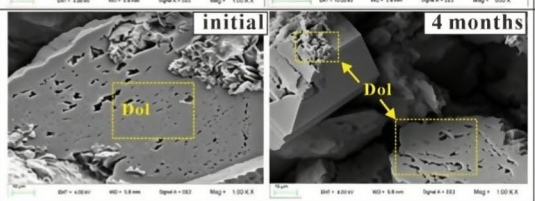
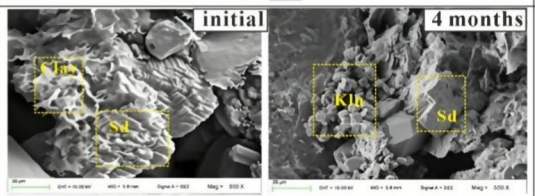
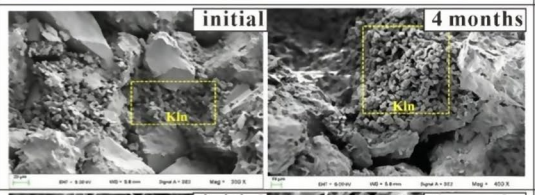
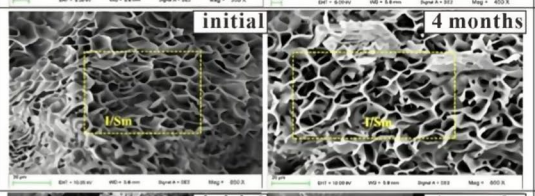
Minrals		SEM images showing changes in mineral occurrence before and after CO_2 dissolution	Minerals changes after CO_2 dissolution
Feldspar	K-feldspar(Kfs)		**After dissolution:** The crystal surface is severely etched by acidic fluid, making the originally smooth surface rough. The cleavage lines are dissolved and expanded, forming dissolution grooves and stepped dissolution surfaces that develop along the cleavage planes.
	Albite(Ab)		**After dissolution:** The elongated crystalline shape is still partially retained, but the crystal faces have been eroded and modified, and the laminated double crystal patterns have been eroded and expanded; there are bay-shaped eroded gaps at the crystal edges, and secondary clay minerals are attached to the surface.
Carbonate	Calcite(Cal)		**After dissolution:** The crystal surface undergoes intense dissolution, making the originally smooth surface rough, the edges rounded, and numerous irregular dissolution pits appearing within the crystal; the intercrystalline cementing substances are dissolved, significantly increasing the porosity between the particles.
	Dolomite(Dol)		**After dissolution:** The micropores are enlarged and interconnected under the action of acidic fluids. The crystal surfaces are etched to form dense dissolution cavities, and dissolution grooves appear in some areas. The degree of dissolution is weaker than that of calcite.
	Siderite(Sd)		**After dissolution:** The crystals underwent intense dissolution and alteration, with their surfaces being eroded and transformed. Some siderite also underwent clayification alteration, converting into clay minerals such as kaolinite (Kln).
Clay	Kaolinite(Kln)		**After dissolution:** The degree of crystal aggregation increases; in low-content areas, dissolution releases pores, while in high-content areas, agglomeration intensifies and throat channels are blocked.
	Illite/Smectite (I/Sm)		**After dissolution:** The honeycomb structure is broken, the micropores increase, but the connectivity deteriorates.
	Chlorite/Smectite (Ch/Sm)		**After dissolution:** The sheet-like structure has been damaged, with some areas experiencing dissolution and the release of space, but overall, it is still mainly clogged.

Fig. 13 SEM images illustrating the morphological changes in typical sandstone minerals before and after 4 months of CO_2-brine-rock interaction, categorized by mineral type: feldspar (K-feldspar, albite), carbonate (calcite, dolomite, siderite), and clay minerals (kaolinite, illite/smectite, chlorite/smectite). The right column summarizes the key dissolution and alteration features of each mineral after reaction [5]

Table 1 Comparison of experimental studies on the micro-mechanisms of CO_2–rock interactions and their sequestration effectiveness

Rock Type	Research Method	Reaction Mechanism	Storage Effect	Ref.
Basalt	^{14}C tracing technique, mineralogical characterization, reaction rate monitoring	Rapid dissolution of Mg- and Fe-rich silicates, stable precipitation of carbonate minerals	Over 95% CO_2 mineralization and immobilization within 2 years; reaction rate initially increases then decreases and stabilizes, consistent with dual-exponential kinetics, achieving permanent storage	[96]
Shale	Microscopic morphology observation, quantitative mineral analysis, pore structure testing	Supercritical CO_2 forms acidic fluid, inducing mineral dissolution	Intense dissolution of carbonate minerals, development of large pores and micro-fractures; substantial improvement in reservoir porosity and permeability, simultaneous consumption of CO_2, achieving long-term geological storage	[98]
Tight sandy conglomerate	Core reaction experiments, pore-throat parameter and petrophysical property testing	CO_2 dissolved in water forms weakly acidic fluid, dissolving feldspar minerals	Feldspar dissolution causes a 0.31% decrease in core mass; dissolution pores develop, pore-throat radius increases, permeability increases by 7.4%–11.0%	[102]
Sandstone	Ion chromatography, SEM, XRD mineral characterization	Acidic fluid drives coupled mineral dissolution–recrystallization reactions	Significant dissolution of calcite, K-feldspar dissolution releases potassium ions; secondary clay mineral migration plugs pore throats, reducing reservoir permeability	[103]
Mudstone	Low-frequency elastic parameter testing, high-frequency ultrasonic detection, microstructural analysis	CO_2-acidified brine dissolves active minerals, modifying clay microstructure	Slight decrease in rock elastic and acoustic parameters; minor mineral dissolution and microcrack propagation; overall mudstone framework remains stable, caprock exhibits good long-term sealing performance	[107]

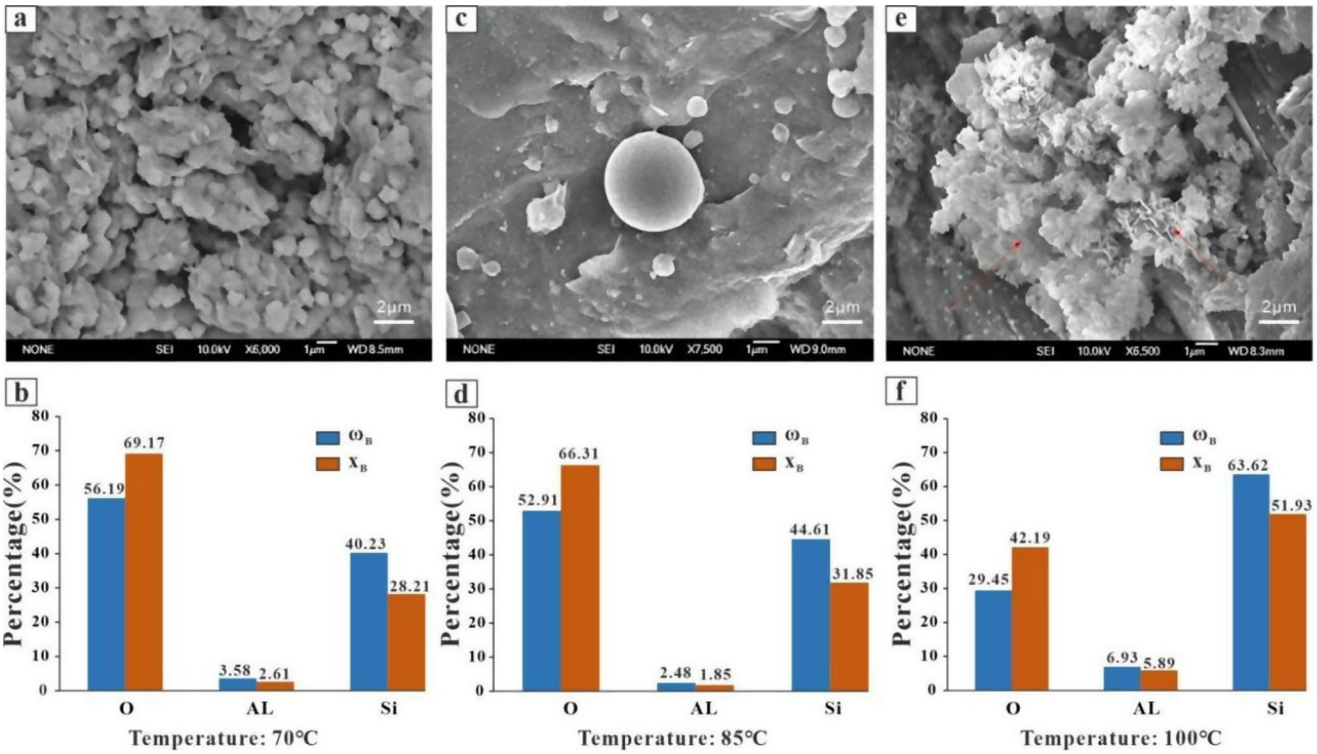


Fig. 14 The formation process of new minerals; (**a**) Acicular minerals formed after reaction at 70°C; (**b**) The mass fraction and atomic fraction of the acicular mineral at 70°C; (**c**) Spherical minerals formed after reaction at 85°C; (**d**) The mass fraction and atomic fraction of the spherical mineral at 85°C; (**e**) Acicular and floral minerals formed after reaction at 100°C; (**f**) The mass fraction and atomic fraction of the acicular and floral mineral at 100°C.(The scanning electron microscope images are taken from the literature, and the data are derived from the literature [110]

rock-mechanics experiments on CO_2 geological sequestration. Using scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDS), they analyzed dissolution and precipitation interfaces in rocks. CO_2 dissolution generates carbonic acid, which induces dissolution of feldspar and calcite, releases metal ions, and promotes secondary crystallization, ultimately altering pore structure and enabling mineral sequestration [115]. Li et al. (2023) built a two-dimensional model using the TOUGHREACT simulator, based on the actual geological conditions of the

Shiboxi Formation in the Ordos Basin, to simulate reaction-driven migration in the CO_2–brine–rock system and the subsequent CO_2 mineralization sequestration. CO_2 dissolution lowers pH, causing dissolution and ion release from primary minerals; these ions then combine to form secondary minerals, so both porosity and permeability first increase and then decrease. Although the reaction proceeds slowly, this process achieves high sequestration safety [116]. Dai et al. (2024) performed mineral dissolution and precipitation experiments for shale-based CO_2 sequestration, simulating geothermal temperature, pressure, and fluid conditions; XRD and SEM were used to observe mineral evolution. During the short-term reaction phase, calcite Ca–O bonds dissociate and Ca^{2+} is released, while dolomite, quartz, and feldspar remain stable. The process is dominated by selective dissolution by acidic fluids; supercritical CO_2–acidic fluids preferentially dissolve carbonate minerals, accompanied by precipitation of secondary minerals, which modifies the pore-throat structure at the microscale and affects sequestration stability [117].

The microscale mechanisms of CO_2–rock interaction and mineralization sequestration remain disputed, with no consensus on several core issues. Mineralization efficiency varies widely across rock types: basalt exceeds 95% mineralization within two years, whereas sandstone or shale shows an initial increase followed by a decrease in permeability because of dissolution–precipitation competition. Systematic cross-rock-type comparisons are still lacking, however, so the fundamental controls on these differences remain unidentified and reservoir selection has no unified mineralogical criterion. Views also diverge on how temperature regulates mineral-evolution pathways—different temperatures produce distinct mineral morphologies—but this finding comes from specific systems and does not generalize to multi-mineral coexistence. Future work should address three areas: building a unified mineralogical evaluation system, clarifying multi-mineral coupled evolution, and developing multi-field coupled, multiscale prediction methods. These efforts would establish quantitative links among rock type, temperature, and evolution pathways, moving CO_2 mineralization sequestration from descriptive observation to quantitative prediction and engineering-adaptability assessment.

5 Challenges and prospects

5.1 Challenges

Research on the microscopic oil-water-rock reaction mechanisms of CO_2 flooding and storage has advanced from macroscopic displacement characterization to nano- and micro-scale interfacial analysis. Nevertheless, several core scientific bottlenecks still restrict the large-scale, long-term and safe engineering application of CCUS technologies, which are summarized as follows.

1. Ambiguous coupling relationships of multiple microscopic mechanisms and lack of quantitative criteria. Current studies mostly focus on individual mechanisms such as CO_2 viscosity reduction, swelling extraction, competitive adsorption, and mineral dissolution, while the synergistic and restrictive coupling relationships among these multiple mechanisms remain unclear. The coupling regulation laws of pore confinement, wettability, mineral properties and other factors are unspecified. In addition, there is a lack of quantitative criteria for the primary-secondary transformation mechanisms and critical conditions of various microscopic effects, making it difficult to accurately characterize the complex displacement processes in reservoirs.
2. Controversial dual regulatory mechanism of formation water and unclear action principle. Formation water presents dual properties of blocking mass transfer and constructing seepage channels. Existing studies have not reached a unified conclusion on the critical transformation conditions and main controlling factors of the water film “barrier-channel” effect. The competitive relationship between the positive effect of water-phase seepage enhancement and the negative effect of permeability resistance cannot be effectively decoupled, leading to insufficient understanding of the CO_2 displacement regulation mechanism in water-bearing reservoirs.
3. Strong working condition dependence of interfacial confinement effects and insufficient full-cycle dynamic mechanism research. The superiority of CO_2 displacement and adsorption is significantly restricted by pore size, temperature, pressure, mineral type and other conditions, and no universal evolutionary law has been established. Most existing studies concentrate on the positive gas injection displacement process, while ignoring negative effects such as heavy component re-adsorption and pore throat blockage during the pressure reduction stage. Research on the oil recovery attenuation mechanism throughout the full injection-production lifecycle is deficient, and the ideal models poorly match the actual heterogeneous reservoirs.
4. Cross-scale disconnection between microscopic mechanisms and macroscopic engineering indicators. Most current microscopic simulations and physical experiments are carried out under ideal conditions, divorced from the actual reservoir environments characterized by high temperature, high pressure, high salinity and strong heterogeneity. There is a lack of quantitative correlation

between microscopic reaction mechanisms and macroscopic development indicators including oil recovery factor, gas channeling risk and sequestration stability, which fails to guide the optimization of field injection-production parameters effectively.

5.2 Prospects

Aiming at the aforementioned core scientific bottlenecks restricting the development of CO_2 flooding and carbon sequestration technology, the future research can be carried out from the following four key directions to complement and improve the theoretical system of CO_2-EOR and carbon sequestration, and support the safe and large-scale engineering application of CCUS technology.

1. Quantitative characterization of multi-mechanism coupling and improvement of microscopic theoretical system. This direction focuses on the synergistic coupling of multiple microscopic effects. It quantifies the primary-secondary relationships and transformation thresholds of core mechanisms including dissolution, extraction, adsorption and mass transfer, establishes quantitative criteria for microscopic displacement under multi-factor coupling, and improves the multi-field coupled theoretical system of CO_2 flooding.
2. Improvement of full-cycle dynamic displacement mechanism oriented to real reservoir conditions. Based on actual reservoir geological characteristics, it establishes composite pore and multi-component fluid models. It systematically reveals the full-cycle dynamic evolution of adsorption-desorption behaviors, quantifies the displacement boundary under diverse reservoir conditions, clarifies the long-term productivity attenuation mechanism, and optimizes the interfacial displacement theory.
3. Quantification of competitive water-rock reactions and establishment of long-term reservoir evolution prediction methods. It defines the critical transformation conditions of dissolution-precipitation competition in multi-mineral systems, reveals the evolutionary laws of reservoir pore-permeability and mechanical properties under different working conditions, and builds a long-term prediction system balancing enhanced oil recovery and carbon sequestration safety.
4. Construction of cross-scale correlation models and promotion of theoretical engineering application. Previous studies have systematically expounded the microscopic reaction mechanisms of carbon dioxide in the process of oil displacement, including its interactions with crude oil, formation water, and reservoir rocks. There is a quantitative correlation between these different microscopic reaction mechanisms and the macroscopic characteristics of reservoir development (Table 2), which has significant guiding significance for on-site engineering implementation. In the future, research should be conducted from multiple perspectives such as microscopic simulation, physical experiments, and reservoir numerical simulation to establish a quantitative cross-scale correlation between the microscopic mechanisms and macroscopic development indicators. A comprehensive technical system covering mechanism analysis, parameter optimization, and on-site application should be established to promote the large-scale on-site application of the collaborative technology of carbon dioxide flooding and carbon sequestration.

6 Conclusions

CO_2 flooding and geological sequestration are pivotal technologies that couple enhanced oil recovery with carbon emission reduction. Their application performance and long-term stability highly depend on the microscale physicochemical interactions of the oil-water-rock multiphase system. CO_2 improves crude oil fluidity and strips adsorbed oil films from rock surfaces through dissolution, swelling, viscosity reduction, light component extraction and miscible effects. Clarifying the microscopic conditions of CO_2-induced viscosity reduction and miscible extraction provides a molecular-level basis for determining the minimum miscibility pressure and optimizing injection parameters. CO_2 reacts with formation water to form a weak carbonic acid system, which provides an acidic environment for water-rock reactions and breaks water film barriers to mobilize residual oil in blind pores. Meanwhile, the aqueous phase suppresses CO_2 gas channeling and stabilizes the displacement front via pore occupation, Jamin effect and mobility regulation, achieving full-pore oil mobilization and uniform seepage. In the coupled CO_2-water-rock system, selective mineral dissolution, ion migration, secondary mineral precipitation and wettability alteration jointly reshape the reservoir pore-throat structure. The dissolution of feldspar and carbonate minerals expands pore space and enhances permeability, while the generation and migration of clay minerals may trigger pore-throat blockage. Systematic clarification of the competitive dissolution-precipitation kinetics and secondary mineral evolution provides critical references for predicting reservoir physical property evolution and caprock integrity during long-term carbon sequestration. Accurate revelation and precise regulation of microscale CO_2-oil-water-rock reaction mechanisms serve as the core scientific foundation for improving CO_2-EOR efficiency and ensuring carbon sequestration safety, possessing important theoretical and engineering significance

Table 2 Quantitative relationship between the micro-mechanisms of CO_2 flooding and macro-engineering indicators

Microscopic mechanism	Applicable Reservoir Conditions	Quantitative relationship between microscopic mechanisms and macroscopic engineering indicators	Technology Maturity	Ref.
Viscosity reduction	1.medium-low permeability sandstone reservoir 2.medium-shallow conventional light oil sandstone reservoir 3.medium-shallow conventional sandstone reservoir 4.nanopore reservoirs of shale and tight oil	1. With the continuous dissolution of CO_2, the viscosity of crude oil decreases from 24.8 mPa·s to 2.0 mPa·s, which enhances the fluidity of the oil phase. Meanwhile, the residual oil saturation decreases from 24.33% to 23.00%, which directly promotes the synchronous improvement of sweep efficiency. 2. At 323 K, as the CO_2 dissolution increases, the expansion coefficient rises from 1.00 to 1.73, leading to a significant reduction in crude oil viscosity. 3. Adding 0.1 wt% of SiO_2 nanoparticles can increase the solubility of CO_2 by 17%, further reduce the viscosity of crude oil by 12.4%, and increase the microscopic sweep coefficient from 51% to 68%, while the crude oil recovery rate increases by 9% to 13%. 4. The viscosity of pure crude oil (OMP = 100) is 0.80 mPa·s. After injecting CO_2, when OMP = 50 is reached, the viscosity drops to 0.18 mPa·s.	1.Laboratory and small-scale tests 2.Laboratory 3.Laboratory 4.Laboratory	[21], [23], [29], [31]
Extraction	1.low-permeability continental sandstone reservoir 2.continental shale oil reservoir	1. The displacement pressure was increased from 5.6 MPa to 20 MPa to achieve a mixed phase. The IFT of oil and gas decreased from 16.907×10^{-3} mN/m to 3.49 mN/m, and the proportion of C_{20} light components extracted by CO_2 reached 67.5%. The recovery rate reached up to 90%. 2. Low-speed $scCO_2$ removes oil adsorbed on the wall through hydrogen bonding and extracts light hydrocarbons. The utilization of residual oil in the nano blind end increased by 27%, and compared with high-speed injection, the recovery rate increased by 11% to 16%.	1.Laboratory and small-scale tests 2.Laboratory	[35], [36]
Miscibility	continental tight sandstone oil reservoir	When the displacement pressure is less than 22.6 MPa, it is a non-miscible condition. Before the CO_2 breakthrough, the recovery rate is 37.09%, and after the CO_2 breakthrough, the recovery rate only increases by 2.01%; when the pressure is between 22.6 and 27.9 MPa, it is a near-miscible range. Before the CO_2 breakthrough, the recovery rate is 42.51%, and the recovery rate by molecular diffusion mass transfer increases by 6.79%; when the pressure is ≥ 27.9 MPa, it reaches a completely miscible state, and the final recovery rate can reach 86.50%.	Laboratory and small-scale tests	[42]
Water film rupture	1.medium-shallow conventional homogeneous sandstone oil reservoir 2.continental siliceous shale oil reservoir	1.Although CO_2 flooding does not have the Jamison resistance caused by continuous water film formation, gas migration can occur with a pressure difference of only 0.1 to 0.2 MPa. However, the critical production pressure difference for residual oil at the blind end of the small hole still reaches 2.2 to 2.5 MPa. If CO_2-WAG water-gas alternation is adopted, the critical production pressure difference for residual oil can be reduced by 0.8 to 1.1 MPa, with a reduction rate of 30% to 37%. 2. Under the conditions of 383 K, inlet pressure of 32 MPa, outlet pressure of 30 MPa, and a pressure difference of 2 MPa, a 2500 ps non-equilibrium molecular simulation was carried out. The process was divided into three stages: diffusion from 0 to 181 ps, rupture of the water bridge from 181 to 1125 ps, and pressure difference-driven oil expulsion after 1125 ps. These stages successively involved CO_2 infiltration, reduction of the density in the middle of the water bridge to 0 to form a continuous channel, which caused the expansion of crude oil and enhanced its fluidity.	1.Laboratory 2.Laboratory	[67], [70]
Dissolution	1.tight sandstone oil reservoir 2.shale oil reservoir 3.conventional freshwater sandstone reservoir	1. Acidic CO_2 dissolves rock minerals, creating a large number of interconnected secondary pores. When CO_2 is continuously injected, the permeability increases by 48.77%. 2. In the case of sufficient formation water, carbonate dissolution creates pores, with the permeability increasing by an average of 775.35%; in the absence of water, the fluid carrying capacity is extremely poor, and the mineral debris produced by dissolution cannot be washed away by the fluid, blocking the pores, resulting in an average decrease in permeability of 59.83%. 3. After 0 to 6 months of CO_2 injection, the pore throats of basalt widen by 1.5 to 3 times, and the permeability increases from 4.05 mD to 16.2 mD. The improvement in pore connectivity increases the recovery rate by 9% to 16%; after 2 years of filling, the pore throats are filled with 20% to 50% of iron dolomite, resulting in a 45% to 80% decrease in permeability and a 13% to 19% decrease in recovery rate.	1.Laboratory and small-scale tests 2.Laboratory 3.Laboratory	[5], [98], [110]

Note: OMP denotes Oil Mass Percentage; IFT denotes Interfacial Tension

for the large-scale application of CCUS technologies and the realization of carbon neutrality and sustainable energy development.

Author contributions Yupeng Qiao: Conceptualization, investigation, formal analysis. Yiming Han: investigation, Conceptualization, writing-original draft preparation. Bin Deng: investigation, Yinmiao Kong: data curation, visualization. Haoxiang Ma: formal analysis. Runliang Fang: investigation.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Ethical approval Not applicable.

Competing interests The authors declare no competing interests.

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